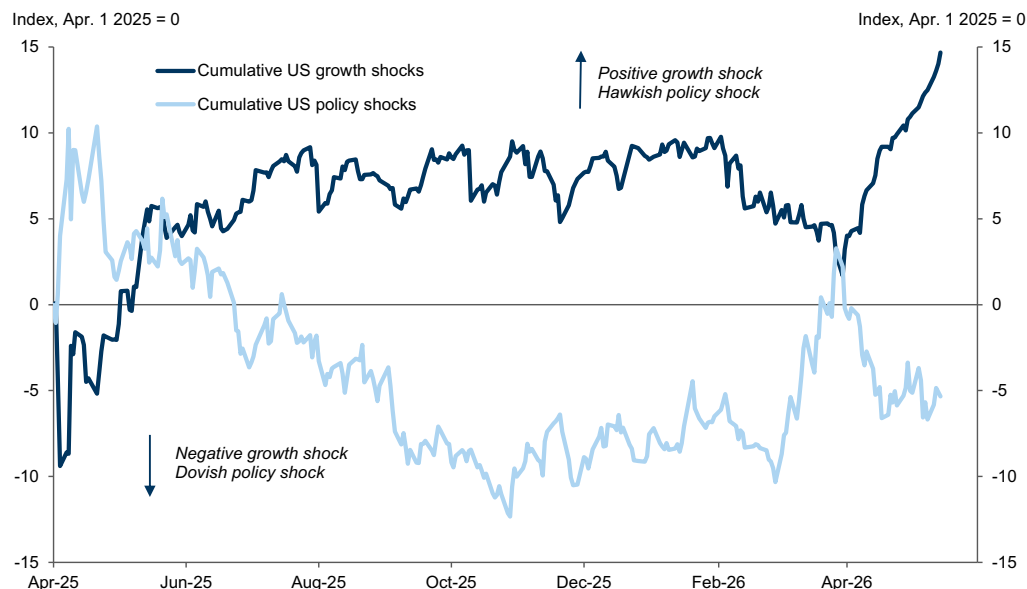


Global Market Views: Shortages

1. The risks of relief. The Iran ceasefire and the reduction in deep downside tails have allowed the market to look ahead and compress risk premia across a range of assets. That type of narrowing in the distribution is often associated with sharp market gains in situations of fundamental uncertainty, and this has proved to be the case again. Despite oil prices and yields staying elevated, a broad range of assets—including US equities, high carry/commodity FX, and EM—have recovered smartly, and renewed AI optimism has pushed those exposures to new cycle highs. The common element is that shortages—in commodities and the AI supply chain—are driving capital flows. After a sharp recovery, the distribution of risks is more balanced. The most obvious downside threat is still a re-escalation of hostilities in the Middle East or a prolonged Hormuz closure that raises the probability of even higher oil prices, higher rates, lower growth and recession. As we wait, the flip side of shortages alongside more resilient growth is upward pressure on bond yields, which the market tends to worry about more when they break new ground. By contrast, even a gradual restart of energy flows should allow some relief across oil prices and rates markets that are now quite hawkishly priced, and lead to a broadening of the positive price action geographically and across sectors. Beyond that, we are in many ways back to the tensions we highlighted in our [outlook](#)—between well-valued assets and increasing concentration on the one hand (with AI-related exposures a poster-child for that), and a macro cycle that does not yet show the imbalances and leverage excess that is typical of late-cycle behaviour. Those tensions mean that even if we avoid the worst tails, we are likely to see volatility rise alongside further equity price increases.

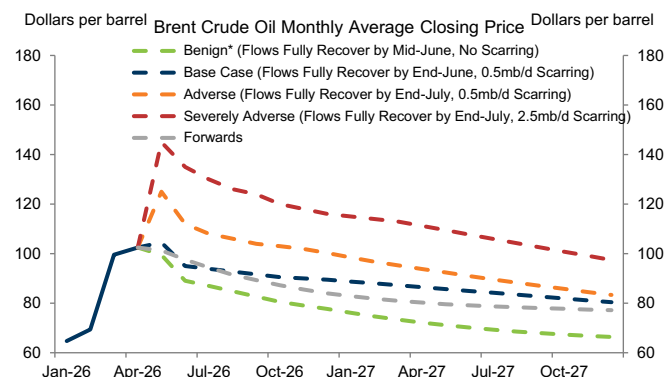
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Exhibit 1: Sharp relief in growth worries, policy shock lingers

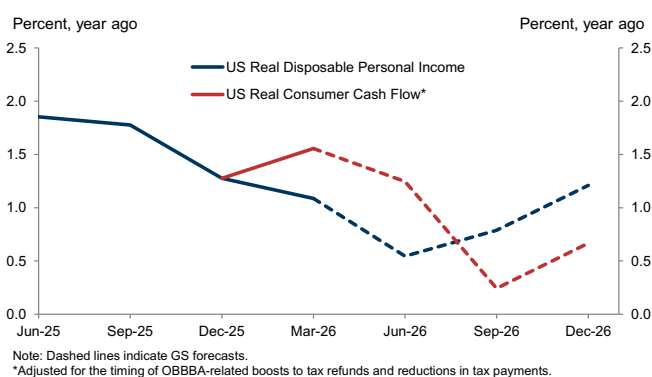
Source: Goldman Sachs Global Investment Research

2. Iran re-escalation still the main risk. The biggest risk to the macro picture still comes from a failure to resolve the Iran war and the flow of oil. Equities and other risk assets have priced in substantial relief on the basis that the deep tail risks are avoided. While that assumption holds, markets have been more tolerant of headline risk. But the flip side to this higher threshold for worry is that a more serious challenge to the market's optimism could lead to a larger repricing. Although we agree that the risks have fallen, the prospect of more adverse outcomes is still very real, and we think that deeper downside tail is underpriced. There is a challenging circularity here too: although markets are looking through any temporary disruptions, it may also be that another bout of market worry is needed to force an agreement that allows oil flows to resume. The longer we go without a clear peace agreement and a convincing reopening of the Strait of Hormuz, the more likely we are to revisit that risk as energy product shortages become clearer. We continue to think there is value in hedges against this outcome. Out-of-the money downside in European equities, credit and FX still screens best in our cross-asset comparisons. Longs in oil may also continue to be protective, though are also vulnerable to a full resolution of the crisis.

Exhibit 2: Oil price risks still skewed to the upside

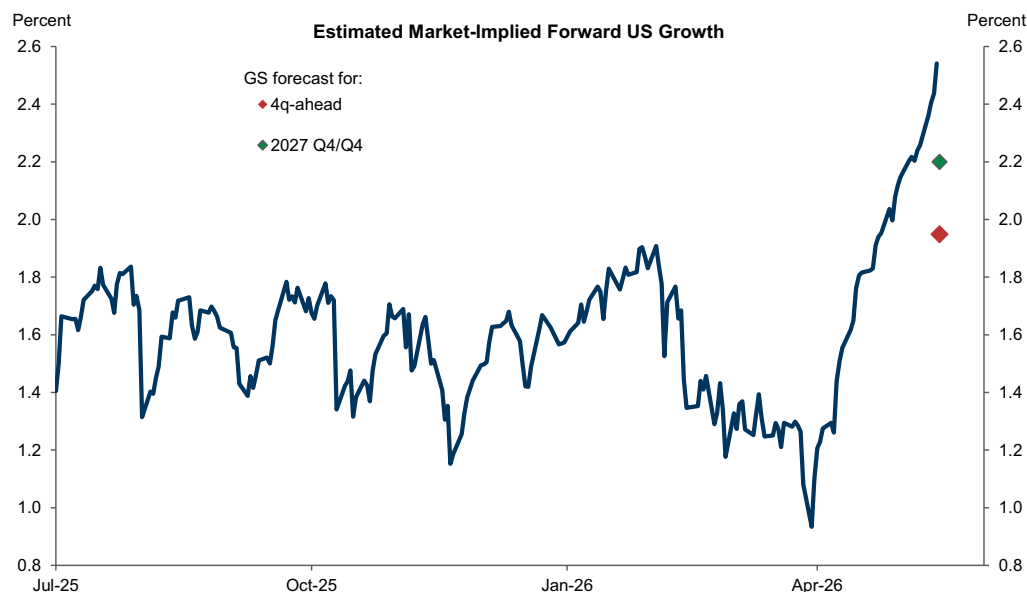
*Benign scenario includes higher US and core OPEC production.

Source: Goldman Sachs Global Investment Research

Exhibit 3: Real income softness ahead for the US

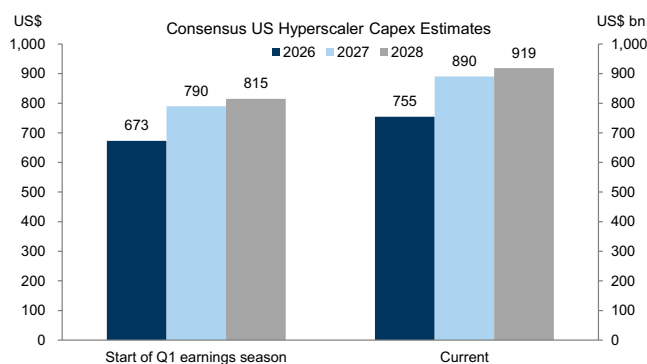
Source: Bureau of Economic Analysis, Haver Analytics, Goldman Sachs Global Investment Research

3. Macro optimism already well priced. The market has already priced significant relief. Our standard estimates of US growth pricing from the joint shifts in equities and bonds now stand at 2.5%. These measures may overstate the true cyclical optimism, since tech and commodity strength may be boosting them for other reasons, but they are consistent with a market that is already looking through the short-term weakness we expect to the better growth outlook in 2027, and is now potentially overshooting those levels. On our forecasts, there is room for some policy relief in the months ahead (our rate forecasts are generally dovish to market pricing) conditional on an Iran resolution. But with the US economy and labor market holding up better than widely expected, and inflation likely to be elevated at least in the short term, the scope for near-term policy relaxation is likely limited unless there is a very clear improvement in oil flows and an end to the war. Part of the reason why our measures of growth pricing have risen sharply is that bond yields and equities have been rising steadily together. Markets are now beginning to question whether that combination is sustainable. We think that inflation pressure is likely to begin to fade within a couple of months of the energy price peak and the market is already priced for hikes, including now in the US. So, while we may well see further near-term pressure for higher yields, we think that pressure will ultimately be contained. But given how relaxed equity markets have been about rising rates so far, markets may continue to worry in the coming weeks that a mix of more hawkish policy pricing and less bullish growth pricing may be needed.

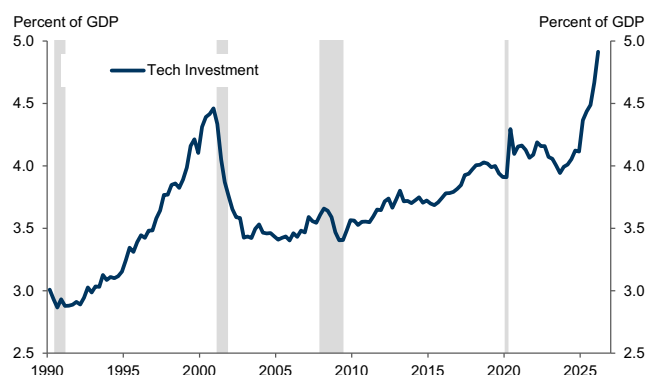
Exhibit 4: Market pricing of US growth already moving past our 2027 growth forecasts

Source: Goldman Sachs Global Investment Research

4. Back to AI amid booming capex. The AI theme is back in force and has again been a focal point for markets. Outside energy-related areas, the markets that have moved most decisively through their pre-war highs are AI-heavy indices such as Korea, Taiwan and Nasdaq. As in 2025, concerns in Q1 about the sustainability of the AI boom have dissipated in Q2 in a way that has overlapped with, but is separate to, a broader macro shock. The value creation here continues to favor heavily those involved in supplying the AI investment boom, especially in the semis/memory spaces where shortages are clear. Capex projections have moved sharply higher, driving positive earnings revisions across those beneficiaries, and tech investment spending has now exceeded the peaks from the late 1990s as a share of GDP. But with profits holding up (corporate profits as a share of GDP are also at their highest on record), most of the macro imbalances that signaled growing trouble in the late 1990s are still not visible yet. Worries about private credit, which we think are unlikely to have systemic consequences, have also settled a little.

Exhibit 5: Another sharp rise in hyperscaler capex expectations

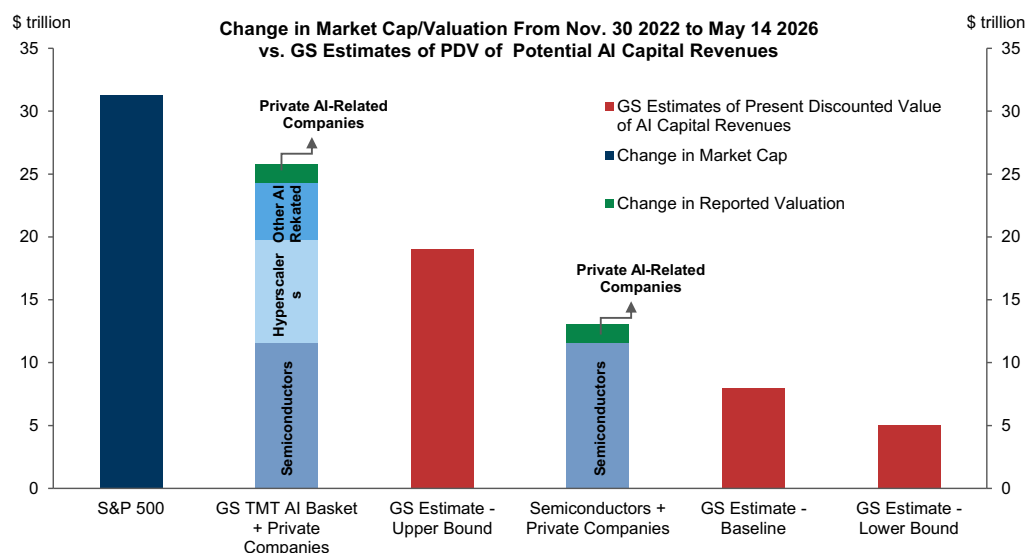
Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 6: Tech investment already at record levels

Shading indicates NBER recessions. Tech investment includes private nonresidential fixed investment in software and information processing equipment.

Source: Haver Analytics, Goldman Sachs Global Investment Research

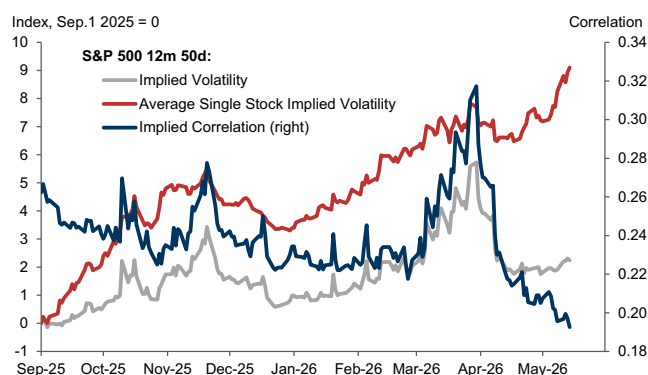
5. The value challenge. As we highlighted in our outlook for 2026, the AI theme is a poster child for the broader tension between a macro cycle that (ex the Iran shock) still looks like it has room to run and market pricing that already looks quite generous. The latest rally has continued to drive the cumulative value embedded in AI-related companies higher. And while not all of this is directly attributable to AI, the market is taking credit for substantial macro benefits. This increases the vulnerability to worries about whether the benefits will be justified and whether they are appropriately allocated (in particular, whether it is sustainable for them to accrue so heavily to the chip layer). There has been more focus on potential winners and losers this year—and dispersion has risen. But the risk is that the market falls into either a fallacy of aggregation (assuming that more individual companies can be winners than the economy overall can deliver) or of extrapolation (assuming that earnings supported by the investment boom itself can be sustained). As long as earnings and spending plans continue to rise more quickly than expected, there is likely to be a tailwind to the AI complex. But in the process, the market may build a valuation overhang that may need to be resolved at some point.

Exhibit 7: Even more value built into AI-related areas

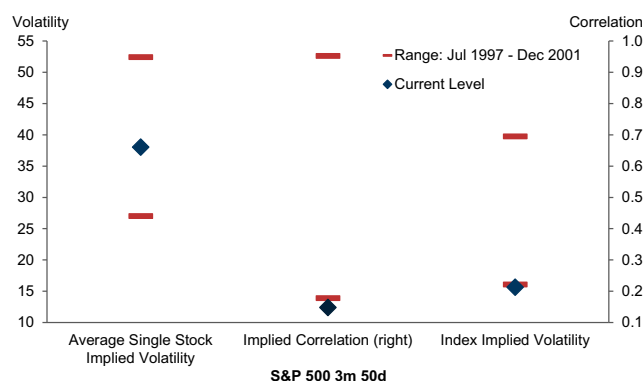
"Semiconductors", "hyperscalers", and "other AI-related" are constituents of the GS TMT AI basket (developed by GS Global Banking & Markets)

Source: Bloomberg, FactSet, Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

6. Higher volatility, but not yet broad-based. One of our core views for 2026 was that longer-dated equity volatility was likely to rise structurally, as volatility around the AI theme and its financing or late-cycle macro dynamics became more prominent. Since last September, there has been a clear upward drift in longer-dated implied volatility in the SPX, despite a sharp rally in equities over that period. But the rise in average single stock volatility—and in concentrated indices like Korea—has been much larger and more persistent than the index story. We think that this is partly because there has been more focus on “distributional” volatility within the AI theme (winners versus losers) than on “macro” volatility (the aggregate value). This has raised single stock volatility but pushed correlations to record lows, limiting the upward pressure on volatility in broad-based indices. We still think that there are good reasons for index volatility to rise over time and any macro shock should drive correlations higher. So we still like pairing long equity views with longs in longer-dated SPX volatility and see longer-dated equity options as appealing ways to limit losses while retaining exposure to further equity upside. And we generally like the idea of owning volatility across assets, whenever it cheapens meaningfully. But the mix of high single stock volatility and low correlation—which we saw at points in the late 1990s too—does mean that index volatility may rise less than expected in some of these shocks.

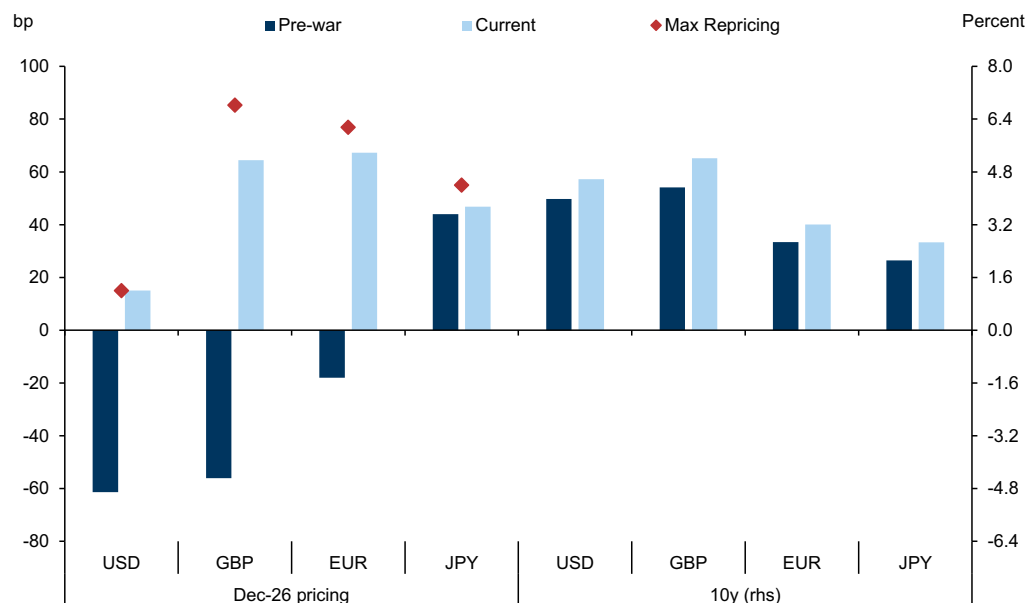
Exhibit 8: Rise in single stock volatility more persistent than at the index level

Source: Goldman Sachs FICC and Equities, Goldman Sachs

Exhibit 9: Index volatility and correlations still low versus the late 1990s

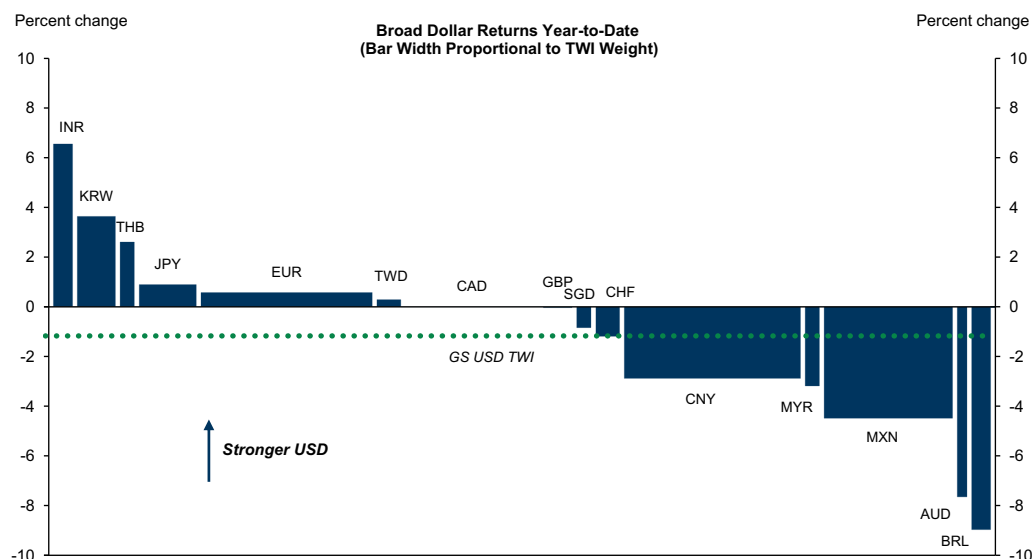
Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

7. Rates running out of time. While many assets have retraced fully and beyond, the hawkish shifts across rates market pricing have generally held. As we move through this messy stalemate where the Strait of Hormuz is still closed, energy prices are still high, growth is holding up, but the first signs of higher prices are coming through in inflation prints, it is possible that the market re-tests or surpasses some of the highs in front-end rate pricing. We continue to think that the threshold for hikes in the US is still very high, or, for that matter, extended hikes elsewhere. But the bulge in inflation without labor market weakening clearly makes easing harder too. Certainly relative to our dovish views coming into the year, we now expect fewer or no cuts across the world, with markets having moved even more hawkishly relative to that. This leaves the forward outlook more balanced than before the war, with more risk of hikes reflected in market pricing. If the energy supply situation resolves, and if the sequential inflation bulge moves through quite quickly, there is again scope for rate relief, particularly front-ends. So, while our bias continues to be that the hikes priced across many DM and EM front-ends are unlikely to be matched, we are also running out of time for cuts in 2026. Linked to that, terminal and longer-dated rates have also been dragged higher by the Iran crisis, and by higher term premia in the UK and Japan. Those moves are also at odds with prior worries about AI displacement and labor market slack, so there too there may be scope for a modest reprieve in the event of an Iran resolution; but it is likely to be limited given ongoing fiscal expenditures (on defense, energy security, or social support) and private spending (on the AI infrastructure build-out).

Exhibit 10: Hawkish repricing has extended from front-ends to long-end

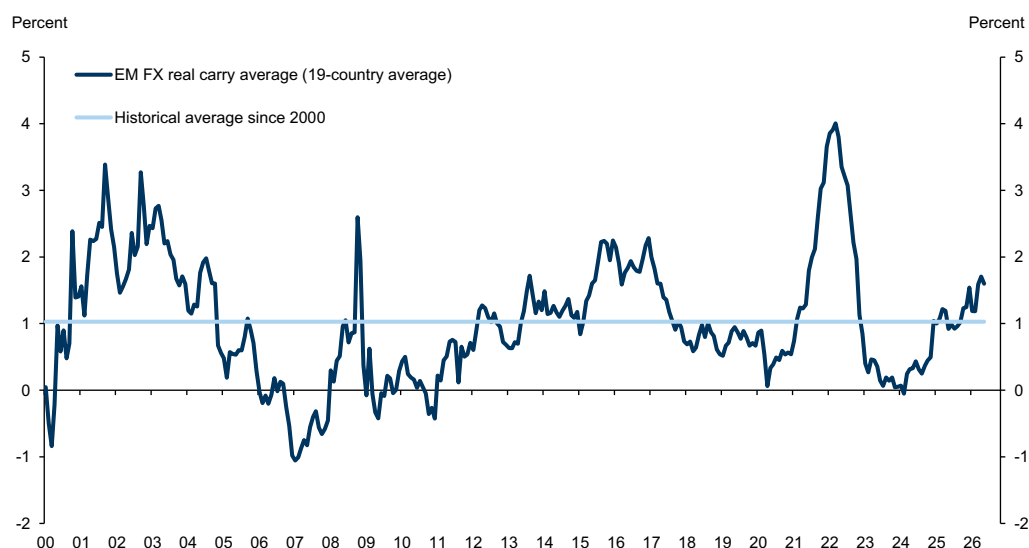
Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

8. Dollar stability masks deeper FX shifts. The reversal in the Dollar since the ceasefire means that it is now largely unchanged, both since the Iran crisis, and year-to-date. Shifts in the Dollar—both up and back down— have been mostly in line with the moves implied by the beta to commodities and risk assets. In that limited sense, Dollar correlations have behaved as expected through this shock so far, and we do not see any new impulse to De-Dollarisation in its broad stability. That stability does, however, mask other significant trends playing out across global FX. Near-neighbour crosses like NOK/SEK (Norwegian versus Swedish Krona) and AUD/NZD (Australian versus New Zealand Dollar) have moved much further to reflect commodity terms-of-trade differences and are now at multi-year extremes. EM FX carry has also continued to generate strong total returns, both versus the Dollar but especially against lower-yielding funders such as the Euro, Swedish Krona and Thai Baht. Potentially even more important is the trend towards sustained CNY appreciation, that has continued through this period. While the Trump-Xi summit could prove a catalyst for more CNY strength, we think that the case for a stronger Renminbi is more fundamental and longer-lasting. China's external surplus is approaching unprecedented levels as a share of global GDP, reflecting deep levels of export competitiveness, and also a highly undervalued currency, with currency appreciation an equilibrium outcome. The Japanese Yen is also very cheap, and while fresh bouts of intervention are a reminder of that reality, we don't see sustained strengthening unless there are stronger concerns about a recession (with lower equities and rates) or there is a more meaningfully hawkish pivot from the BoJ.

Exhibit 11: Broad Dollar stability masks large moves

Source: Bloomberg, Goldman Sachs Global Investment Research

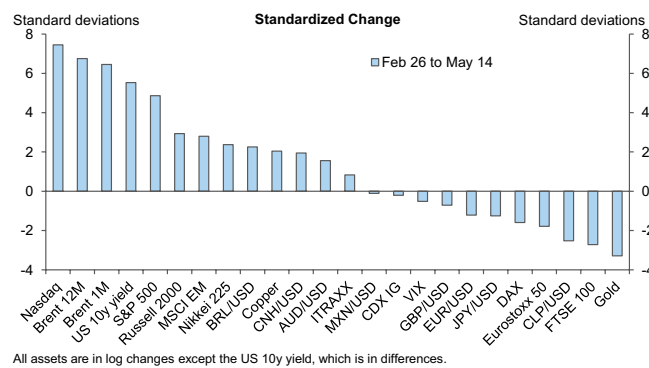
9. EM: Carry on waiting. The EM equity complex, with its exposure to the memory cycle via Korea and Taiwan, and to commodities via LatAm, has managed to generate strong outperformance YTD despite the energy price volatility. That barbell should remain well-suited to the current environment even as the North Asia (long ‘chips’) versus South Asia (short ‘oil consumers’) differentiation becomes even more stretched. But EM carry has been an additional outperformer as we wait in a zone between no clear deal to re-open the Strait of Hormuz but also no kinetic hostilities. Much of this outperformance relates to the fact that many high-carry EM currencies—such as BRL, MXN and COP—also happen to be energy producers. But there has been another underappreciated factor that should lengthen the window in which EM carry can perform. Even if some of the rate hikes priced in DM and EM curves are not delivered, the bar is now higher for rate cuts as Central Banks ‘wait and see’ whether energy supplies are restored, price shocks are indeed transitory, and second-round effects are avoided. As a result, not only is the level of EM carry now higher than it was at the start of the year and high relative to long-term averages, we also now expect a slower decline in real rates and carry levels relative to the start of the year. Taking into account carry levels, relative terms-of-trade shifts and idiosyncratic developments, we prefer BRL, HUF, MXN and ZAR as Longs for an EM Carry Basket. More diversified funders (including EUR, SEK and THB, apart from the Dollar) offer higher carry returns alongside added resilience to energy price and risk sentiment shifts.

Exhibit 12: EM carry has edged higher YTD and is higher than long-term averages

Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

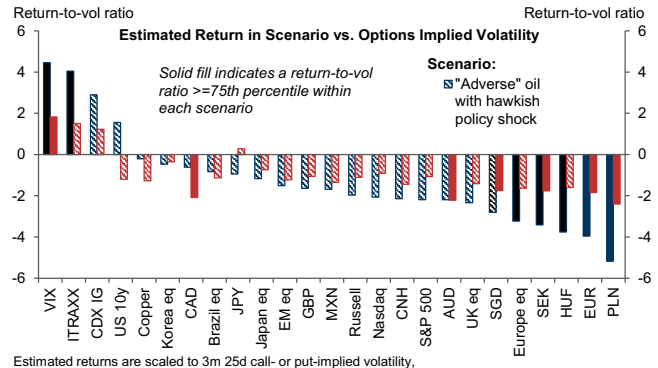
10. Shortages and beyond. Even as the escalatory tail remains crimped, a prolonged messy ceasefire still creates opportunities and risks to navigate. A simple framework for some of the sharpest asset price moves we see is that they reflect physical shortages: prices need to rise sharply to signal that more capital is needed and to incentivize it to flow there. That includes memory chips, power and—at least for now—energy products. Energy resilience is also likely to be a growing theme driving capital spending into renewables across oil-deficient Asia and Europe. Until some of those shortages are resolved, say because new, cheaper sources of supply become available, those moves may be sustained. The counterpart to this is that booming AI investment, fiscal spending, including on defense, and resilient commodity markets may keep bond yields higher than normal. Away from these areas, EM carry strategies can continue to perform against a range-bound Dollar backdrop, and we think it makes sense to retain exposure to the growing divergence between energy exporters/producers and importers/consumers in both equities and currencies as long as full resolution is some way off. In the event of a fuller resolution of the Iran war, the path for risk assets should be further buffered by rate relief (as hike pricing relaxes) and the fact that downside growth tails (from the energy constraint) are set aside. But it is also likely to be punctuated by bouts of volatility given optimistic growth pricing, concentrated positioning, and risks to long-end rates (especially if growth damage is less than feared). So we continue to advocate for meaningful hedges of the downside tails, especially in the shorter term. Renewed worries about credit and equity markets digesting AI-related issuance, returns on those investments, and the aggregation/extrapolation fallacies also argue for pairing constructive risk positions with long volatility exposure to protect against those reassessments and market reversals.

Exhibit 13: Best-performing assets mostly linked to either AI or energy shortages



Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

Exhibit 14: We continue to like hedging for downside tails, with downside in European assets still screening well



Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

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Reg AC

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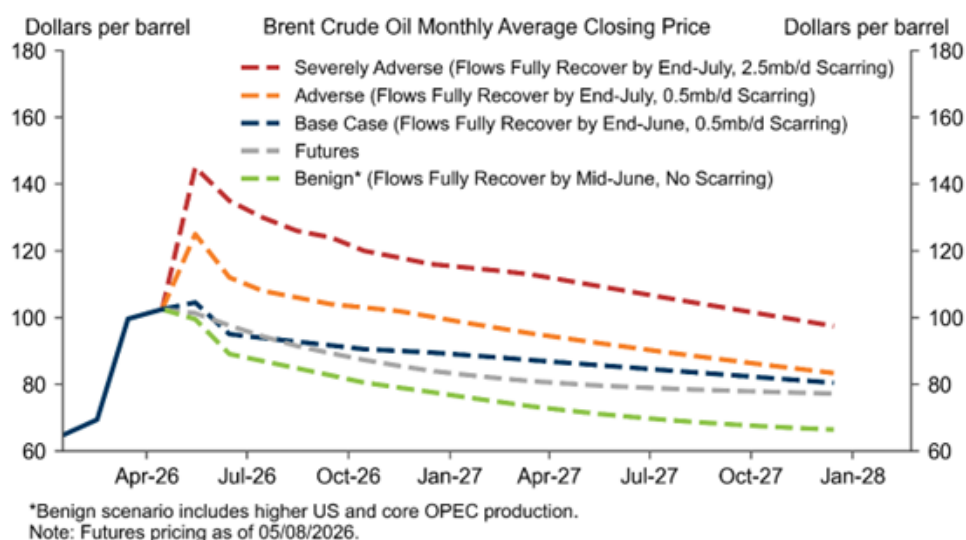
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Global Views: Bending, Not Breaking

1. We see three reasons why the 10-week closure of the Strait of Hormuz has hurt global growth only moderately so far. First, oil prices have not risen as much as expected, partly because of unusually high pre-war inventories and partly because markets never lost faith that very large consumer price hikes would prompt a US policy shift. Second, physical shortages in areas such as jet fuel have so far been met with relatively painless forms of demand destruction (e.g. a large shift to renewables in China and reduced flight schedules on lower-value routes globally). Third, fiscal policy, the AI boom, and—with a brief interruption in March—financial conditions have been supportive all year. Under our baseline assumption of a gradual Strait reopening that starts soon and finishes in late June, we see Brent prices stable in the near term and edging down to \$90/barrel by year-end. However, the risks remain tilted toward more adverse outcomes, higher oil prices, and greater economic damage.

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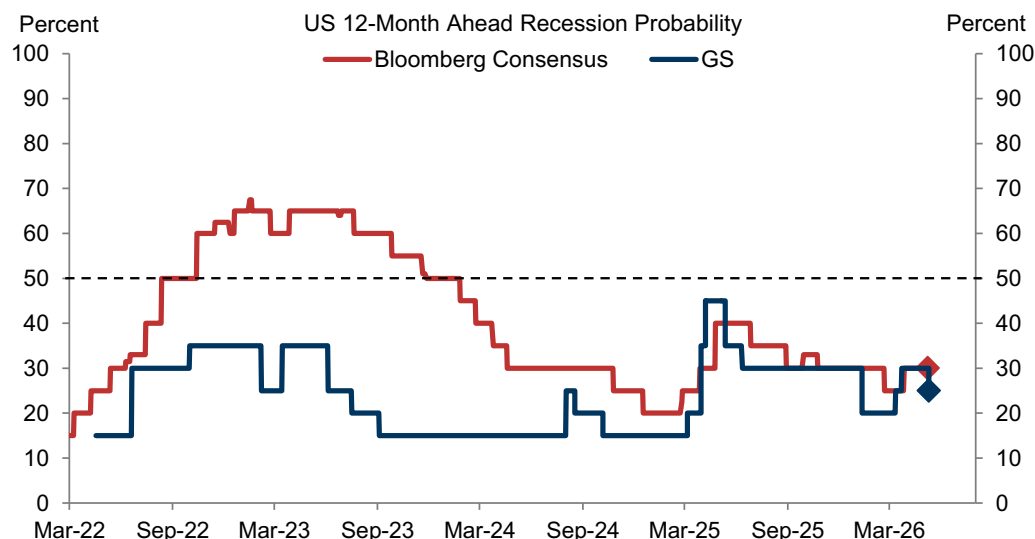
Exhibit 1: Oil Price Risks Still Skewed to the Upside



Source: Goldman Sachs Global Investment Research

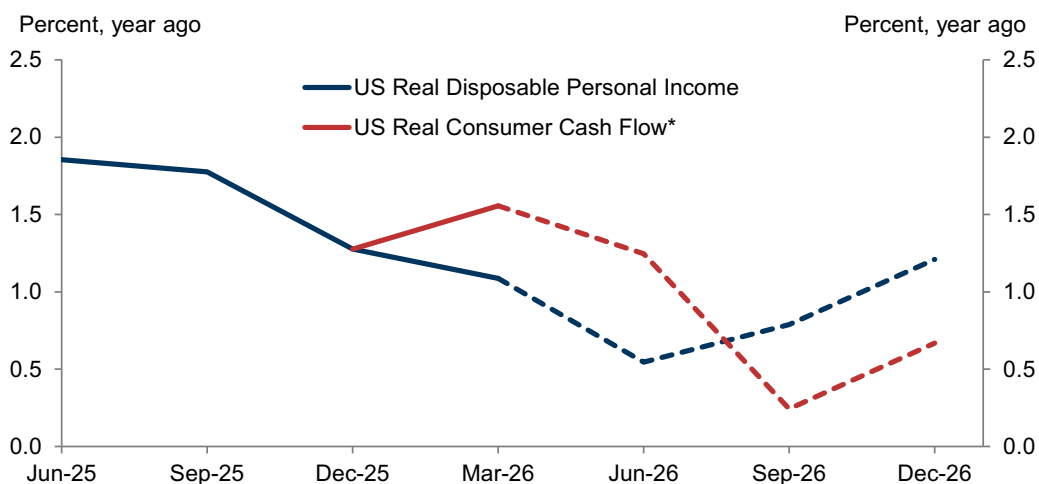
2. We have shaved our 12-month US recession probability from 30% to 25% as economic activity has held up well and our financial conditions index has eased back below pre-war levels. While headline GDP growth fell short of expectations in Q1, private domestic final sales grew at a solid 2.5% in real terms. Moreover, the labor market news has mostly beaten expectations, with a bigger-than-expected 115k increase in nonfarm payrolls in April, a stable 4.3% unemployment rate, and a sharp decline in jobless claims.

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Exhibit 2: Recession Risk Looks Incrementally Lower

Source: Bloomberg, Goldman Sachs Global Investment Research

3. But our recession risk estimate remains 5pp above the pre-war level and 10pp above the unconditional long-term average. Moreover, our baseline forecast entails below-trend growth in coming quarters, mainly because consumers face weakness in real disposable personal income and real cash flow. The boost to cash flow from strong tax refunds is about to fade, and the drags from higher gas prices, slowing nominal wage gains, and reduced healthcare subsidies are likely to persist. With the personal saving rate already down to 3.6%, the lowest level in three years, weaker income and cash flow is likely to translate into weaker consumer spending.

Exhibit 3: Real Income and Cash Flow Are Likely to Soften in H2

Note: Dashed lines indicate GS forecasts.

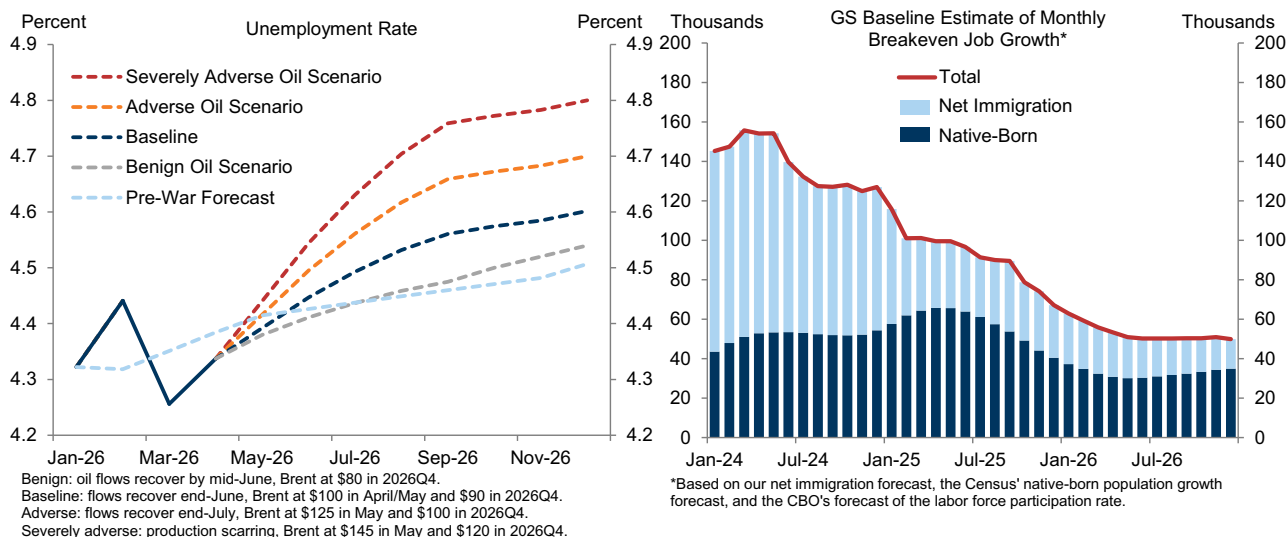
*Adjusted for the timing of OBBBA-related boosts to tax refunds and reductions in tax payments.

Source: Haver Analytics, Bureau of Economic Analysis, Goldman Sachs Global Investment Research

4. Beyond the weaker growth outlook, there are two other reasons why the US unemployment rate is likely to drift higher. First, US productivity has already accelerated from a 1.5% trend before the 2020 pandemic to 2.1% since then, and the improvement

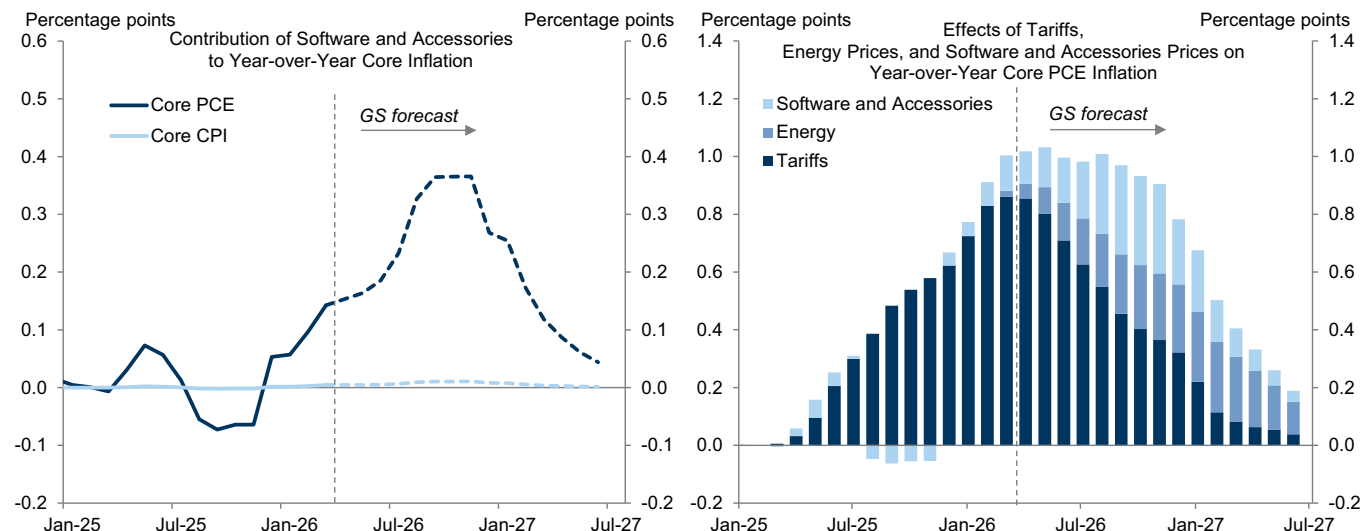
is likely to continue as the AI impact broadens. This means fewer new jobs for any given GDP increase. Second, we believe the Fed staff and many private-sector analysts have overshot in revising down their estimates of the “breakeven” rate of payroll growth to around zero. Our estimate of around 50k is consistent with the fact that the unemployment rate has fallen by only a few basis points on net this year, despite average payroll growth currently reported at 76k.

Exhibit 4: Unemployment Likely to Rise, in Part Because Breakeven Payroll Pace Remains Positive



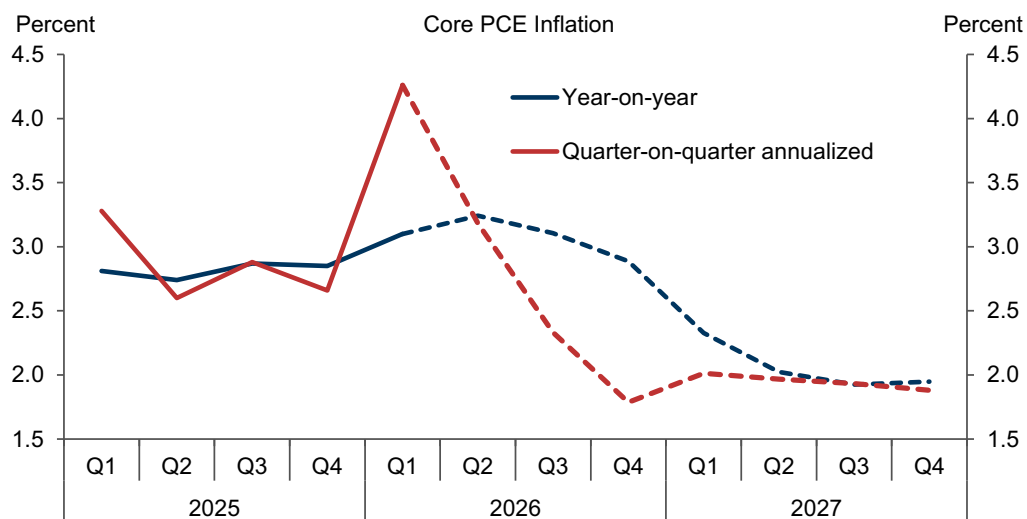
Source: Goldman Sachs Global Investment Research

5. Two distortions related to the AI boom have worsened the near-term outlook for core PCE inflation. First, after adding new AI features to software packages such as MS Office, Microsoft and other firms now charge a higher price for the new bundle than for the prior standalone package. The official statistics don't adjust for the expanded offering and thus incorrectly treat this as a pure price increase. Second, the combined weight of software and accessories—where prices have also surged on the back of costlier memory—is 30x higher in PCE than CPI, apparently because the underlying survey incorrectly captures not only consumer demand but also some demand from small businesses. Together, these distortions are likely to boost the year-on-year rate of core PCE inflation by 0.4pp later this year. Combined with an estimated 0.3pp pass-through from higher energy prices, this will delay the renewed core PCE disinflation we have been expecting.

Exhibit 5: Software Is Yet Another Transitory Inflation Shock

Source: Haver Analytics, Goldman Sachs Global Investment Research

6. The longer-term outlook for core PCE inflation is better. While the boosts from tariffs, energy, and software/accessories should fade over the next year, two key drags are likely to remain in place. First, rent inflation should continue to slow as the official CPI/PCE shelter data catch up with market rents, which remain weak because investors overbuilt apartments in the immediate post-pandemic era in several markets (especially in the Sun Belt). Second, the labor market is a source of disinflation, with nominal wage growth of 3.6% year-on-year based on our GS wage tracker and unit labor costs growing just 1.2% year-on-year in Q1. We therefore see sequential core PCE inflation returning to 2% later in 2026, though the year-on-year rate is unlikely to get back to target until mid-2027.

Exhibit 6: Core PCE Inflation Should Look Better by Year-end, Especially in Sequential Terms

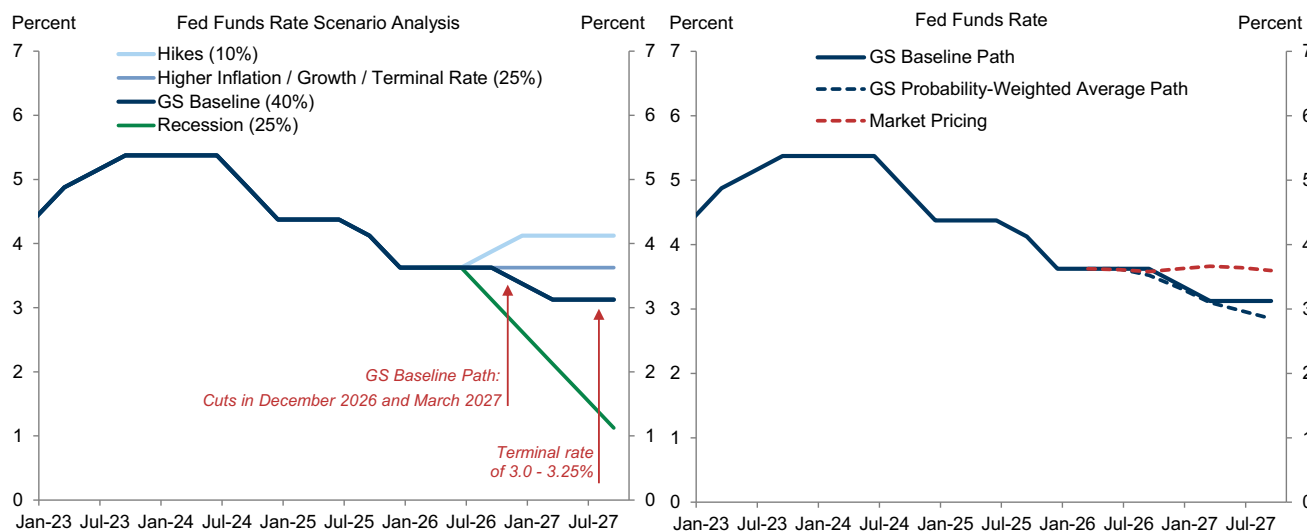
Note: Dashed lines indicate GS forecasts.

Source: Haver Analytics, Goldman Sachs Global Investment Research

7. Upside PCE inflation surprises, better labor market data, and growing hawkishness within the FOMC have led us to delay our forecast for two more 25bp normalization cuts

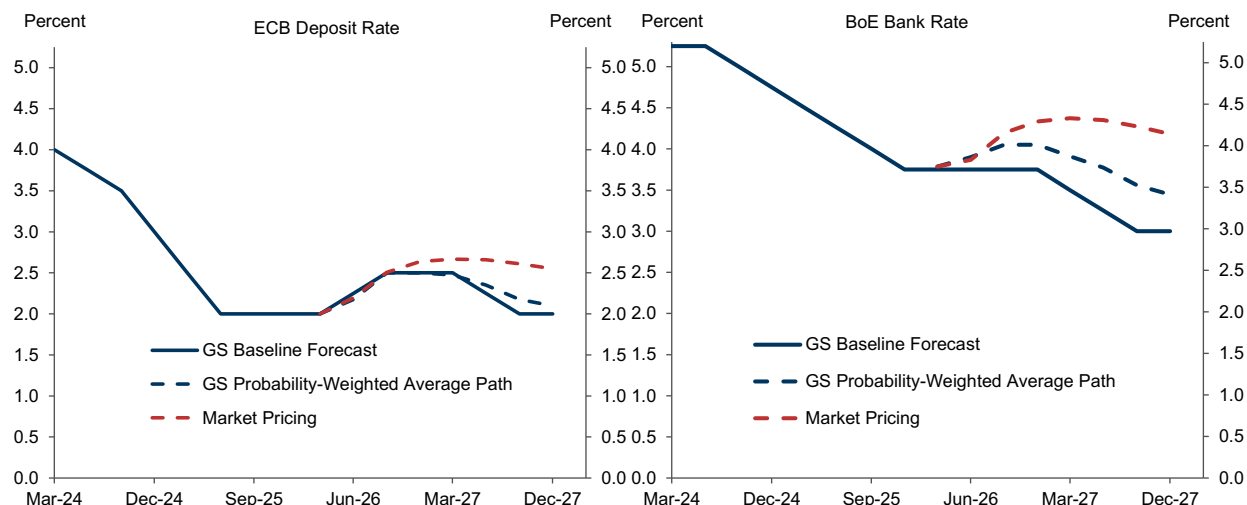
to December 2026 and March 2027. However, our forecast remains below market pricing, especially in probability-weighted terms. One important question will be the Fed leadership transition. We doubt the committee would endorse Kevin Warsh's earlier call for cuts in anticipation of AI-driven disinflation. But his more recent argument that trimmed-mean PCE—which is running 0.8pp below core PCE—is a better measure because it is less sensitive to one-time price shocks seems more promising. After five years of above-target inflation, most sitting Fed officials are reluctant to make this argument publicly, lest they be accused of cherry-picking. But many probably agree with it tacitly and might be persuaded to do so more openly alongside a new Chair.

Exhibit 7: Fed Cuts on a More Delayed Schedule



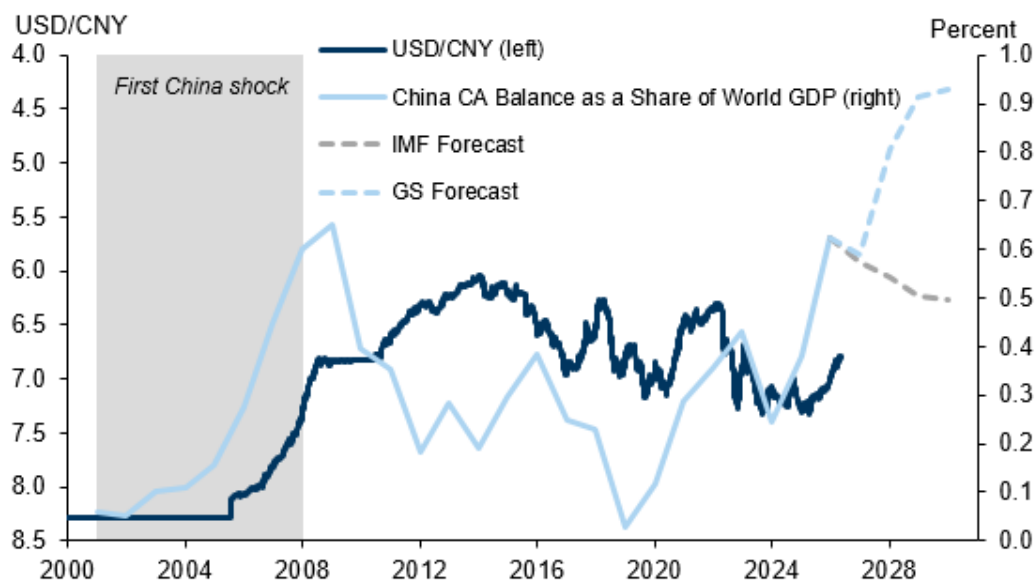
Source: Bloomberg, Goldman Sachs Global Investment Research

8. We still expect the ECB to deliver two 25bp hikes soon—probably in June and September—because of the low 2% starting point for the deposit rate and the historical tendency to (over)react hawkishly to oil-driven headline inflation shocks. However, the risk is on the side of no hikes because core inflation remains very well-behaved so far and ECB officials as a group are more open-minded about core vs. headline than in the pre-Draghi days. In any case, we think that any near-term hikes would probably be unwound in 2027. Across the Channel, we don't see hikes from the BoE in our baseline—in part because Bank Rate at 3.75% is clearly in restrictive territory—but they do constitute a meaningful hawkish risk.

Exhibit 8: We See the ECB Hiking and the BoE on Hold, But Our Views Are Dovish Relative to Market Pricing in Both Cases

Source: Bloomberg, Goldman Sachs Global Investment Research

9. In the run-up to this week's Trump-Xi summit, China's economy continues to benefit from strength in the export-oriented goods sector. This kept Q1 GDP growth at 5.0% year-on-year and our 2026 forecast at 4.7%, but the economy remains very unbalanced. While higher prices for imported energy should temporarily halt the increase in the current account surplus this year, we expect the combination of strong production and sluggish demand to push it further into unprecedented territory as a share of global GDP in subsequent years. The clearest market implication is that CNY is undervalued, especially against the US dollar where our FX strategists estimate a gap of more than 20% and have doubled down on their forecast of significant appreciation, with a new target of 6.50 in 12 months.

Exhibit 9: China's Record Current Account Surplus Should Boost the CNY

Source: Goldman Sachs FICC and Equities, Haver Analytics, IMF, Goldman Sachs Global Investment Research

10. If we wrote down the most frequent topics in discussions with market participants, the vast majority would be negative—an unprecedented oil supply shock, large fiscal

deficits, questions about Fed independence, population shrinkage, cyber/AI security risks, US-China rivalry, and the biggest war in Europe since WW2, amongst others. Moreover, equities are far from cheap. So why have they performed so well? The likely answer is the strength of the Q1 earnings season combined with the improving long-term outlook for productivity and thus profits on the back of the AI revolution. The latter is especially important in the US market, where our strategists estimate that earnings and dividends a decade or more in the future represent roughly 75% of fundamental value. That said, a market where the baseline is positive but the risks are so asymmetrically negative argues for combining long risk position with an exposure to equity volatility or significant hedges (our cross-asset strategists recommend equity puts struck deeply out of the money).

Jan Hatzius

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April 2026: Adoption Rises to 19.8%

- We update our AI adoption tracker through April 2026.
- **AI-related investment growth** remains strong, particularly for semiconductors, where equity analysts expect global revenue growth of 50% from current levels by the end of 2026. AI-related investment in the US national accounts also continues to rise and now stands at \$360bn (1.1% of GDP) above its 2022 level. US investment in AI-related hardware and software continued to rise in today's 2025Q4 GDP release. AI-related hardware shipments from Taiwan to the world rebounded in March and stood at \$61.2bn.
- **AI adoption by firms** rose by 0.9pp to 19.8% among US establishments (with adoption expected to rise to 23.0% in the next 6 months) according to the Census Bureau's Business Trends and Outlook Survey (BTOS). Information services, professional services, education services, and finance/insurance firms continue to lead adoption. The latest BTOS release also included several **new supplemental questions** regarding AI adoption and its impacts on business activity. Sales and marketing, strategy and business development, and R&D are the most common AI use cases among adopters today, and intensity of use of AI tools for these purposes is expected to ramp up sharply in the next 6 months. Most firms are using AI only to supplement a small number of employee work tasks and AI adoption has had a minimal impact on total business employment in most sectors. Among non-adopters, AI incompatibility with the business and a lack of knowledge on AI capabilities were the most commonly cited barriers to adoption.
- **AI's labor market impact** remains narrow. Employment in specific industries like marketing, graphic design, customer service, and especially tech where AI-use cases have been established is contracting by around 11k per month. However, this is offset by construction job growth in data center-exposed categories, which is up 212k since 2022 (relative to broader construction employment trends). The tech share of total employment continues to decline relative to the pre-2022 trend. That said, the correlation between AI adoption and other labor market slack measures remains limited, and only 15.3k employees were affected by corporate layoffs attributed to AI in March (though several other high-profile tech companies have announced layoff plans in the coming months over the last week).
- We continue to observe large **impacts on labor productivity** in the limited areas

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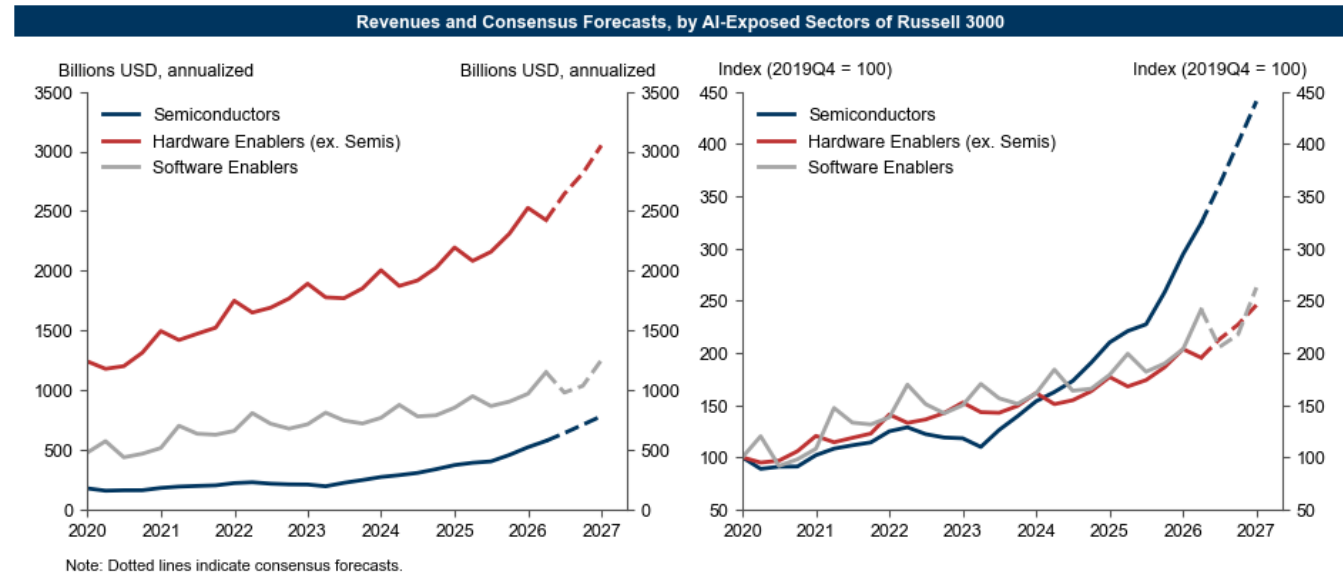
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where generative AI has been deployed. Academic studies imply a 23% average uplift to productivity, while company anecdotes imply slightly larger efficiency gains of around 33%. Industries with higher AI adoption rates are now showing a slight acceleration in productivity growth over the past year in official US data.

April 2026: Adoption Rises to 19.8%

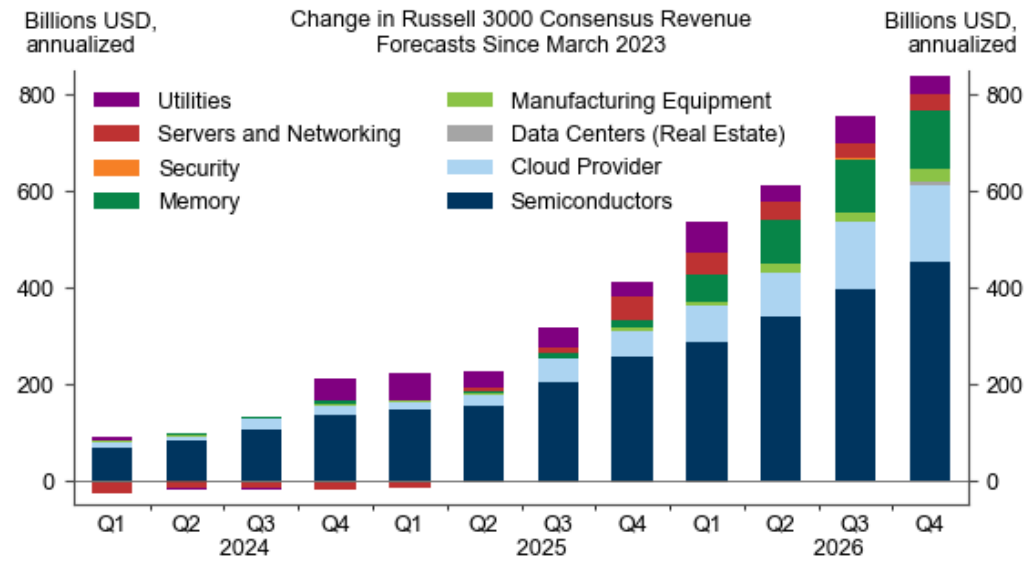
AI Investment

Exhibit 1: Semiconductor Company Revenues Are Expected to Increase by 50% in 2026 on a Q4/Q4 Basis



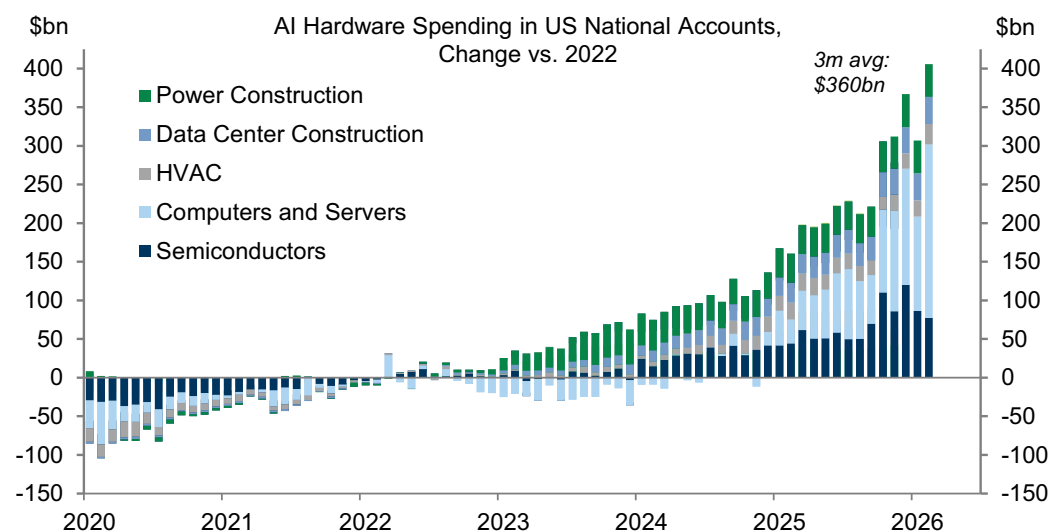
Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 2: Revenue Forecast Revisions Imply that AI-Related Revenues Could Rise to Over \$800bn in 2026Q4



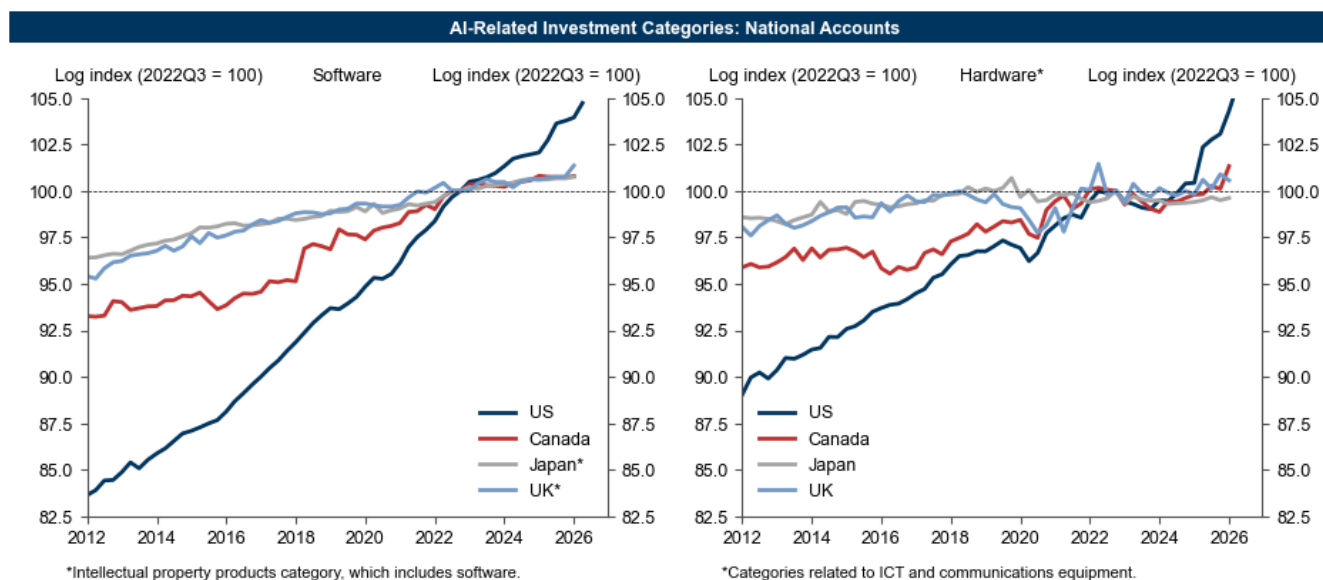
Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 3: AI Hardware Investment in the US National Accounts Has Increased by \$360bn Since 2022 on a Three-Month Annualized Basis



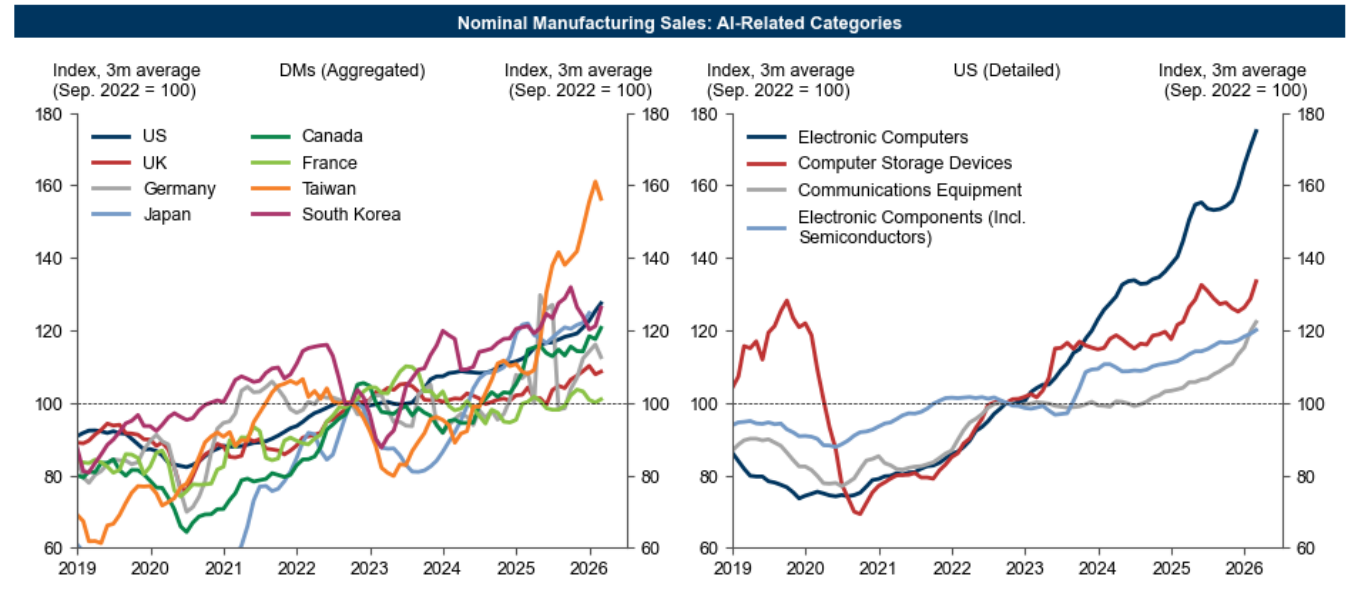
Source: Census Bureau, Bureau of Economic Analysis, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 4: US Investment in AI-Related Hardware and Software Continued to Rise in 2025Q4



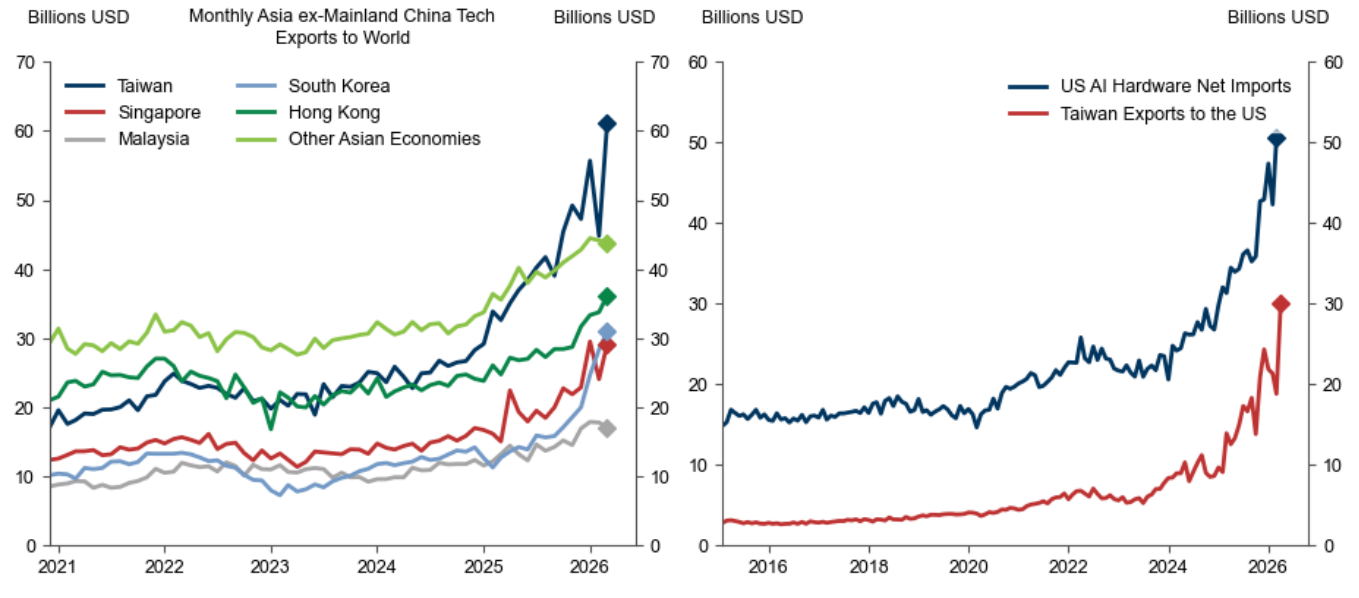
Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 5: AI-Related Manufacturing Shipments Remain Elevated in Taiwan...



Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 6: ...and Taiwan Total Tech Exports to the World Surged to \$61.2bn in March

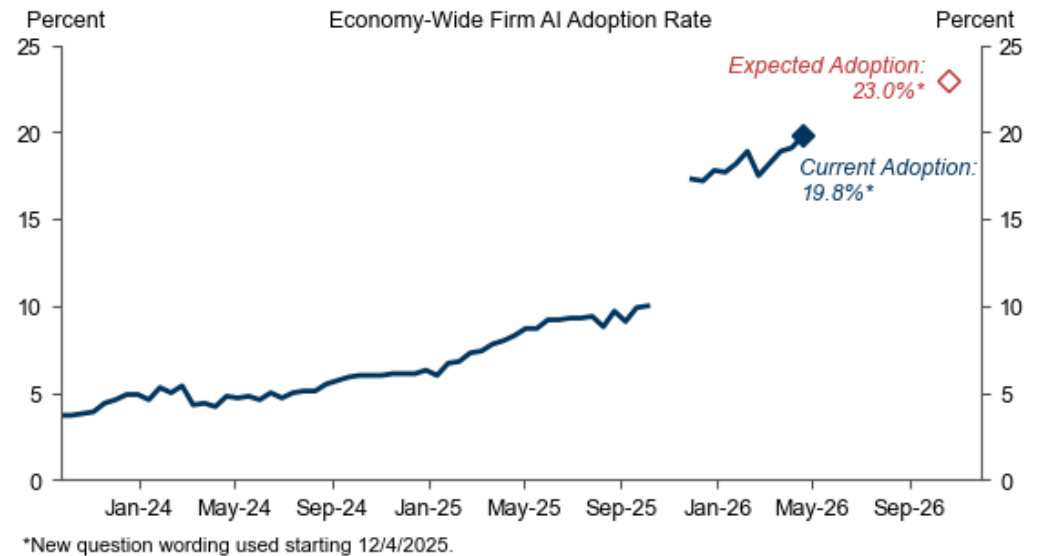


Source: Haver Analytics, Goldman Sachs Global Investment Research

AI Adoption by Firms

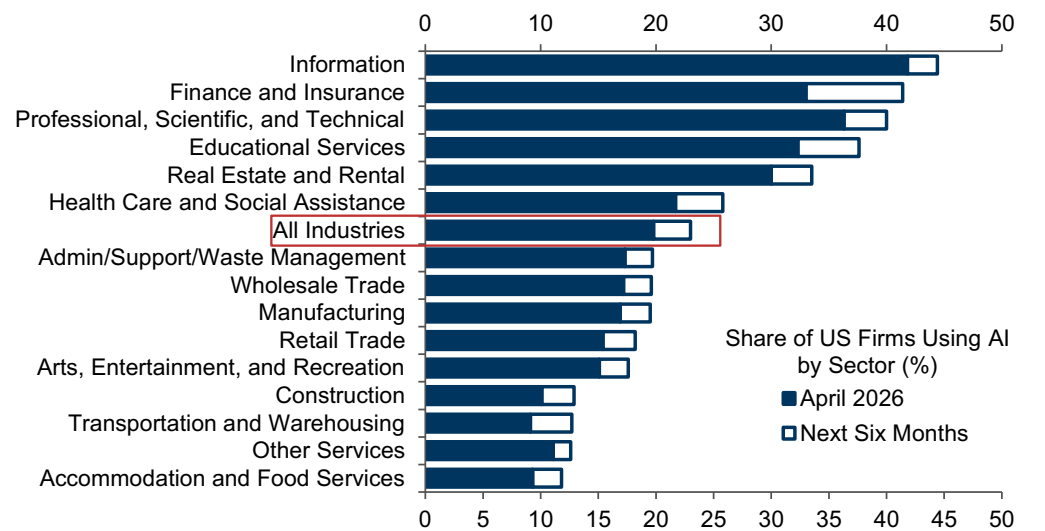
Note: The Census Bureau has rephrased questions from asking about AI use in “producing goods and services” to asking about AI use in “any business functions” starting from the December 4, 2025 release. There was a level shift observed in conjunction with the new question wording, and we will be tracking adoption rates under the new methodology going forward.

Exhibit 7: 19.8% of Firms Are Using Artificial Intelligence in Their Regular Business Functions (+0.9pp From March); 23.0% Expect to Use AI In the Next 6 Months



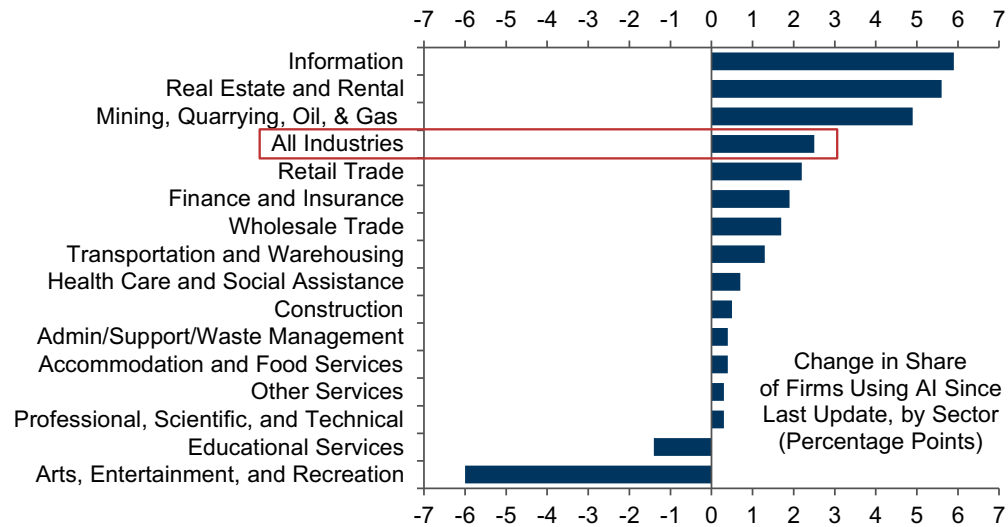
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 8: Information, Professional Services, Education, and Finance and Insurance Establishments Continue to Lead Adoption



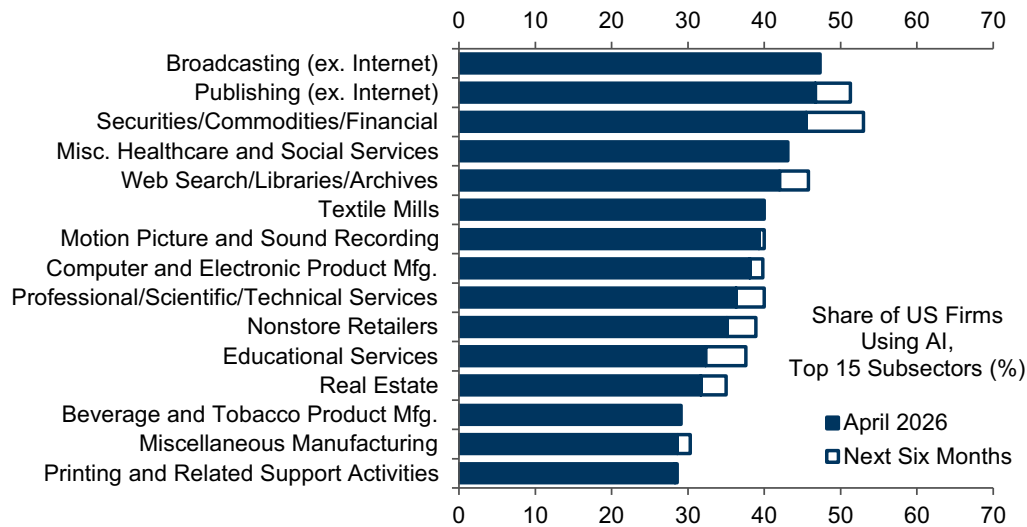
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 9: Information and Real Estate Establishments Had the Largest Gains in Adoption Since Our Last Update



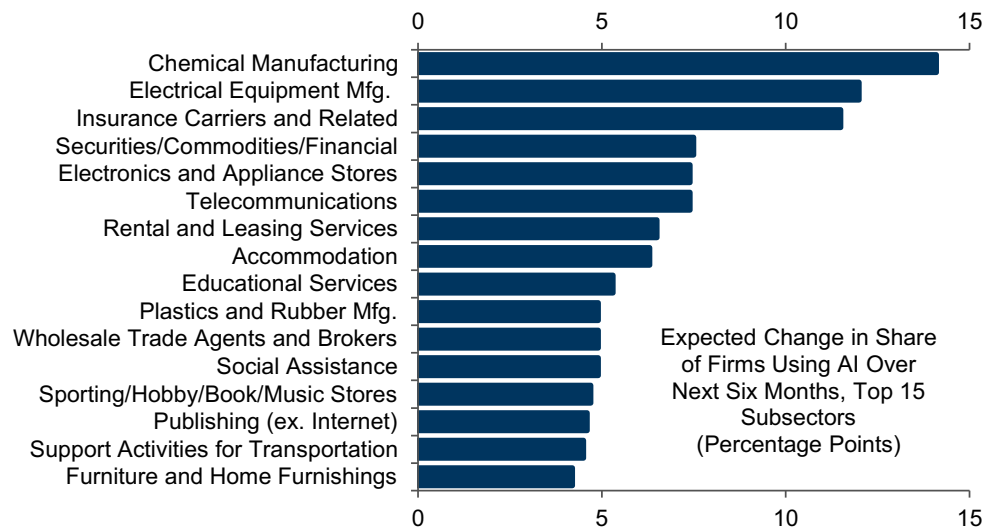
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 10: Among Subsectors, Current Adoption Remains the Highest in Broadcasting, Publishing, and Securities



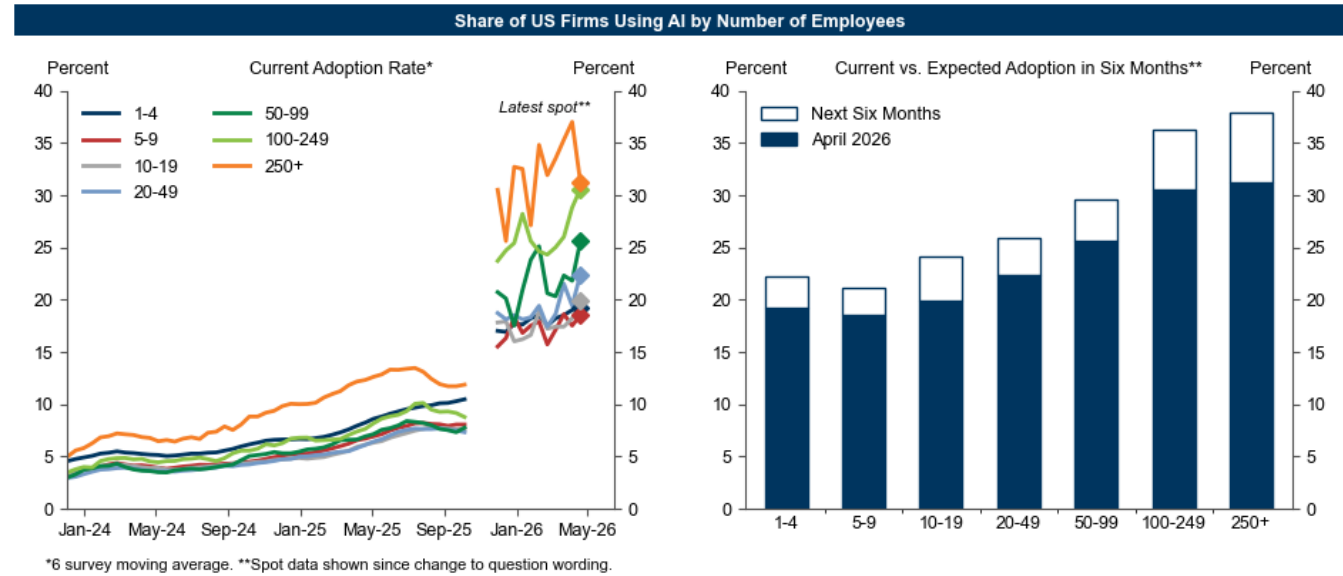
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 11: Chemical and Electrical Equipment Companies Report the Largest Expected Adoption Increases in the Next 6 Months



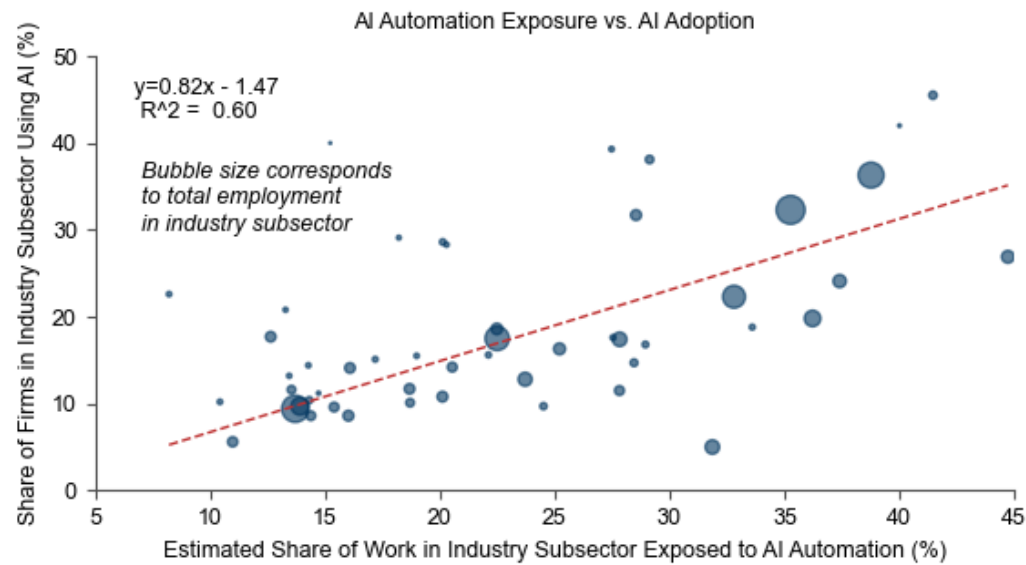
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 12: Current Adoption Among Mid-Sized Firms With 100-249 Employees Rose to 30.6%



Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 13: AI Adoption Remains Strongly Correlated With Our Subsector AI Automation Exposure Scores



Share of work exposed to AI automation is a GS estimate based on the methodology in “Global Economics Analyst: The Potentially Large Effects of Artificial Intelligence on Economic Growth” (March 26, 2023).

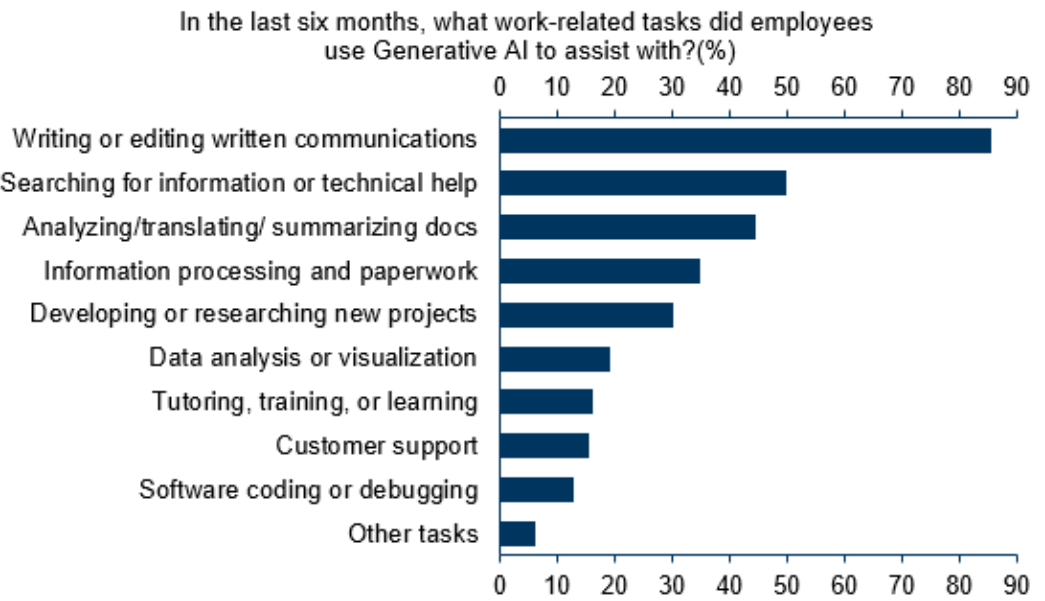
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 14: Sales and Marketing, Strategy and Business Development, and R&D Are the Most Common AI Use Cases...



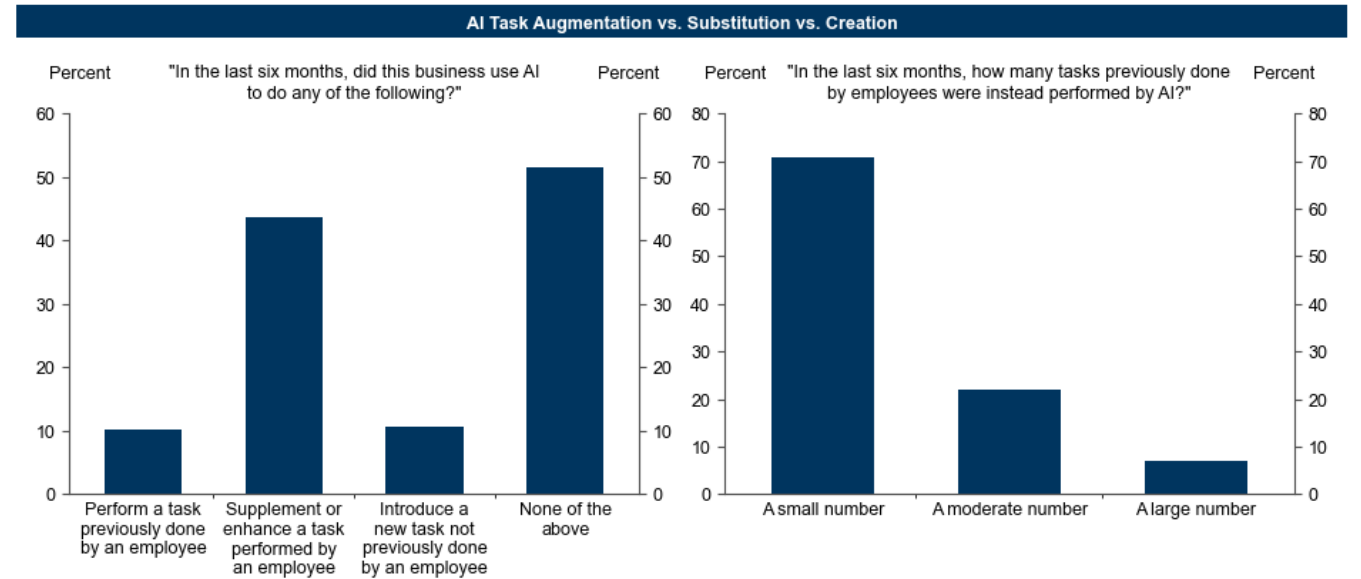
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 15: ...and Writing or Editing Written Communications Was the Most Commonly Cited Work Task that Generative AI Assisted With



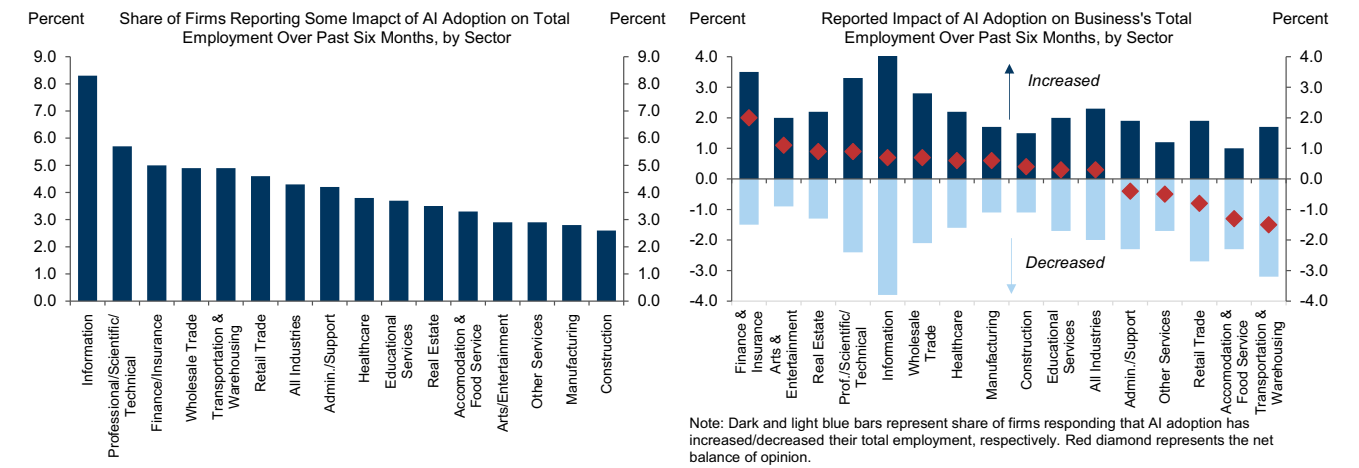
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 16: Most Firms Are Using AI Only to Supplement a Small Number of Employee Work Tasks



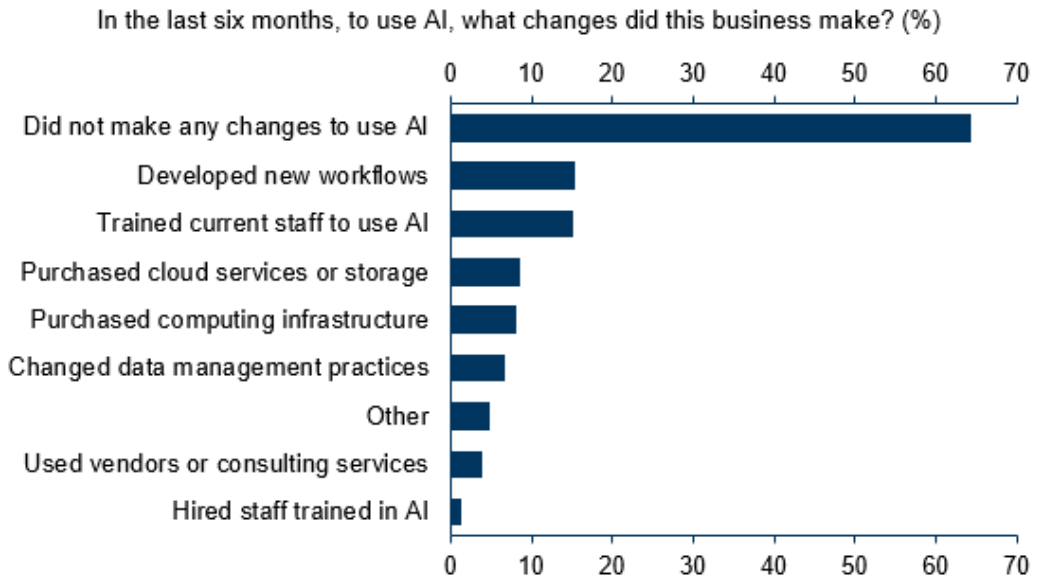
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 17: Information Services, Professional Services, and Finance and Insurance Firms Are Most Likely to Report that AI has Impacted Headcount in Some Way, but the Net Impact on Employment for the Overall Economy Is Small (and Slightly Positive)



Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 18: Most Firms Did Not Make Significant Organizational or Infrastructure Changes to Adopt AI



Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 19: Inapplicability Was the Most Commonly Cited Barrier to AI Adoption Among Non-Adopters



Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 20: A Majority of Surveyed Employees Report Positive Productivity Impacts from Using AI at Work

Source	Date	Current Adoption / Impacts	Future Adoption / Impacts
Gallup	April 2026	Half of all surveyed adults reported using AI in their jobs and 27% of employees in AI-adopting organizations say that their workplace has changed in disruptive ways due to AI in the past year. 65% of adopters said that AI has improved their productivity and efficiency, regardless of how often they personally use AI at work.	Eighteen percent of all U.S. employees say it is very or somewhat likely their job will be eliminated within the next five years due to AI or automation. Among employees working in organizations that have adopted AI, that share rises to 23%. However, employees in AI-adopting organizations were more likely to report both current expansions and reductions in headcount.
NVIDIA	March 2026	64% of respondents said their organizations are actively using AI in their operations. 28% said they're still in the assessment phase, while 8% said they're not using AI and have no plans to. Regional adoption among respondents was 70% in North America, 65% in EMEA, and 63% in APAC. Larger companies (with over 1000 employees) demonstrated broader adoption, deployed more use cases, and reported greater ROI. More than half of respondents (53%) said improved employee productivity was one of the biggest impacts AI had on business operations.	86% of respondents said their AI budgets would increase this year and nearly 40% said budgets would increase by 10% or more. Respondents reported that workflow and operations and retaining the right expertise to scale AI solutions were the key challenges to adoption.
Thomson Reuters	February 2026	40% of respondents said their organizations were using generative AI, up from 22% a year ago. Publicly available tools such as ChatGPT are still the most commonly used platform. In contrast, only 15% of respondents said their organization use agentic AI tools.	53% of respondents said their organizations are either in the planning or consideration phase for agentic AI. 77% of professionals say they expect it to be a central part of their workflow by 2030.
Deloitte	January 2026	In the latest survey, 34% of companies were starting to use AI to deeply transform their businesses, 30% were designing key processes around AI, and the remaining 37% were using AI at the surface level with little or no change to underlying business processes. 84% of companies have not redesignated jobs or work tasks around AI capabilities. 25% of surveyed leaders reported that AI is having a transformative impact on their organizations, more than double from 12% a year ago.	Survey results showed that insufficient employee skills were the largest barrier to integrating AI into existing workflows. Nearly three-quarters of surveyed companies expect to be using agentic AI at least moderately in two years.
Gallup	December 2025	The percentage of US employees who reported using AI at work at least a few times a week increased from 19% in 2025Q2 to 23% in Q3. Unsurprisingly, employees working in knowledge-based jobs (such as tech or professional services) were more likely to use AI than those in frontline positions. Most common use cases include using AI to consolidate information or data, idea generation, and automating basic tasks. Users most often use AI chatbots and virtual assistants.	Gallup research has shown that employee AI adoption is strongly associated with having greater managerial support for AI and strategic integration of AI in their role, indicating that a strategic people-first approach is needed for effective AI adoption.
McKinsey	November 2025	88% of surveyed organizations are regularly using AI in at least one business function, up from 78% last year. However, most surveyed organizations are still in the experimentation / piloting phase of AI and nearly two-thirds have not begun scaling AI across the enterprise. 39% reported experimenting with AI agents specifically and the use of AI agents is most commonly reported among TMT and healthcare businesses. Respondents reported the greatest revenue benefits from AI in marketing and sales, strategy and corporate finance, and product development. A plurality of respondents had little to no change in employee headcount due to AI use over the past year.	32% of respondents expect an overall workforce reduction of 3% or more due to AI use and larger organizations are more likely to expect workforce reductions. Inaccuracy is the AI-related risk that was most commonly experienced and the top priority that organizations are working to mitigate going forward.
Wharton Human-AI Research and GBK Collective	October 2025	Usage of generative AI among enterprise leaders has risen sharply this year and stands at 46% (+17pp yoy). The most common use cases include data analysis, document summarization, and document editing / writing. Nearly three-quarters already see positive ROI on generative AI and four in five expect positive returns within two to three years. Larger enterprises with over \$2B in annual revenue were more likely to report that it is "too early" to measure productivity outcomes of their AI initiatives as they navigate integration complexity.	88% of surveyed business leaders expected an increase in AI spend at their organization in the next 12 months (+16pp yoy). 11% of these leaders are funding AI by cutting spending in other areas, with reallocation most commonly coming from legacy IT and HR programs.
Gartner	June 2025	45% of surveyed firms with high AI maturity reported that their AI initiatives remained in production for 3+ years to ensure sustained impact and value, compared to 20% in low-maturity organizations. 91% of leaders from high-maturity organizations have appointed dedicated AI leaders.	Security threats, finding the right use cases, and data availability / quality were among the top barriers for AI implementation reported.
University of Maryland Center for Global Business	June 2025	58% of the business executives and middle managers surveyed reported that their firms had incorporated AI into their business operations in some capacity. 39% reported using generative AI, 30% reported using computer vision, and 27% reported using machine learning. A majority of respondents said that customer experience and operations efficiency improvements were at the core of their AI adoption strategies	A majority of respondents declared to strongly or somewhat support AI regulations on transparency and/or algorithmic bias. 20% of respondents were very or extremely concerned about AI-driven job displacement in the near future.
Paypal and Reimagine Main Street	June 2025	Out of nearly 1000 small businesses surveyed, 25% reported having already integrated AI into their daily operations while another 51% reported experimenting with AI (but not incorporating into regular operations). Business automation, financial intelligence, and real-time customer insights were cited as some of the most common strategic applications of AI tools.	Respondents who had not yet fully adopted AI cited security concerns, resource constraints, and uncertain value add as the top barriers faced. Most respondents who had not yet adopted AI were in varying stages of engagement on a trajectory towards adoption.

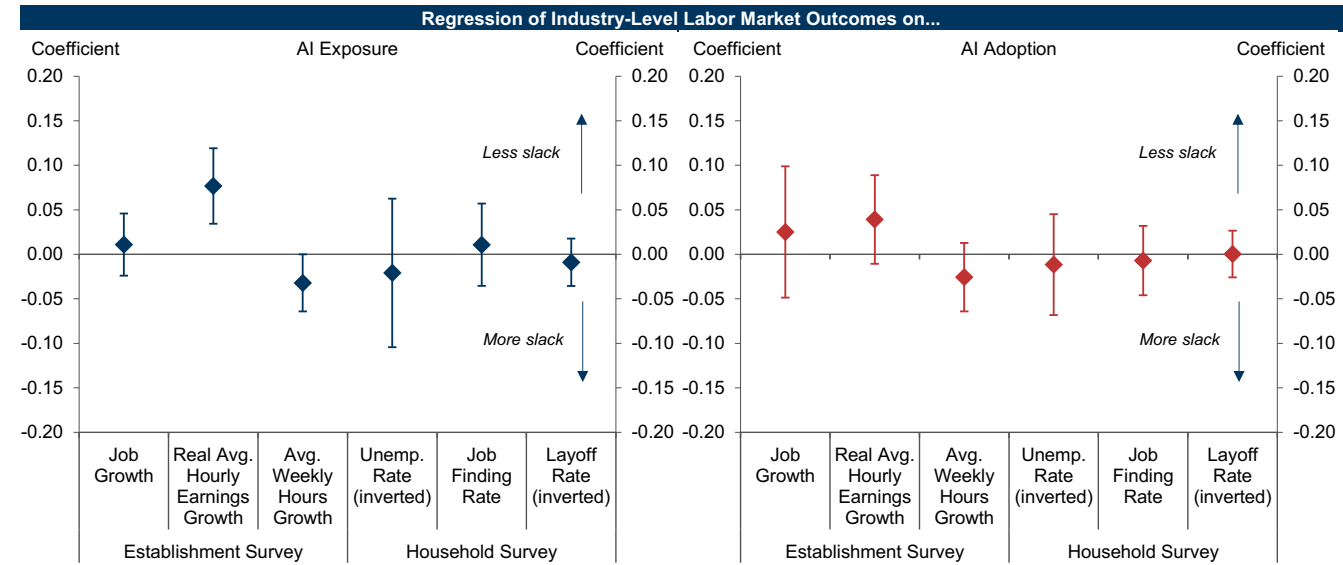
Green shading indicates surveys added since the last quarterly update.

Source: Goldman Sachs Global Investment Research

AI’s Impact on the Labor Market

Labor Outcomes

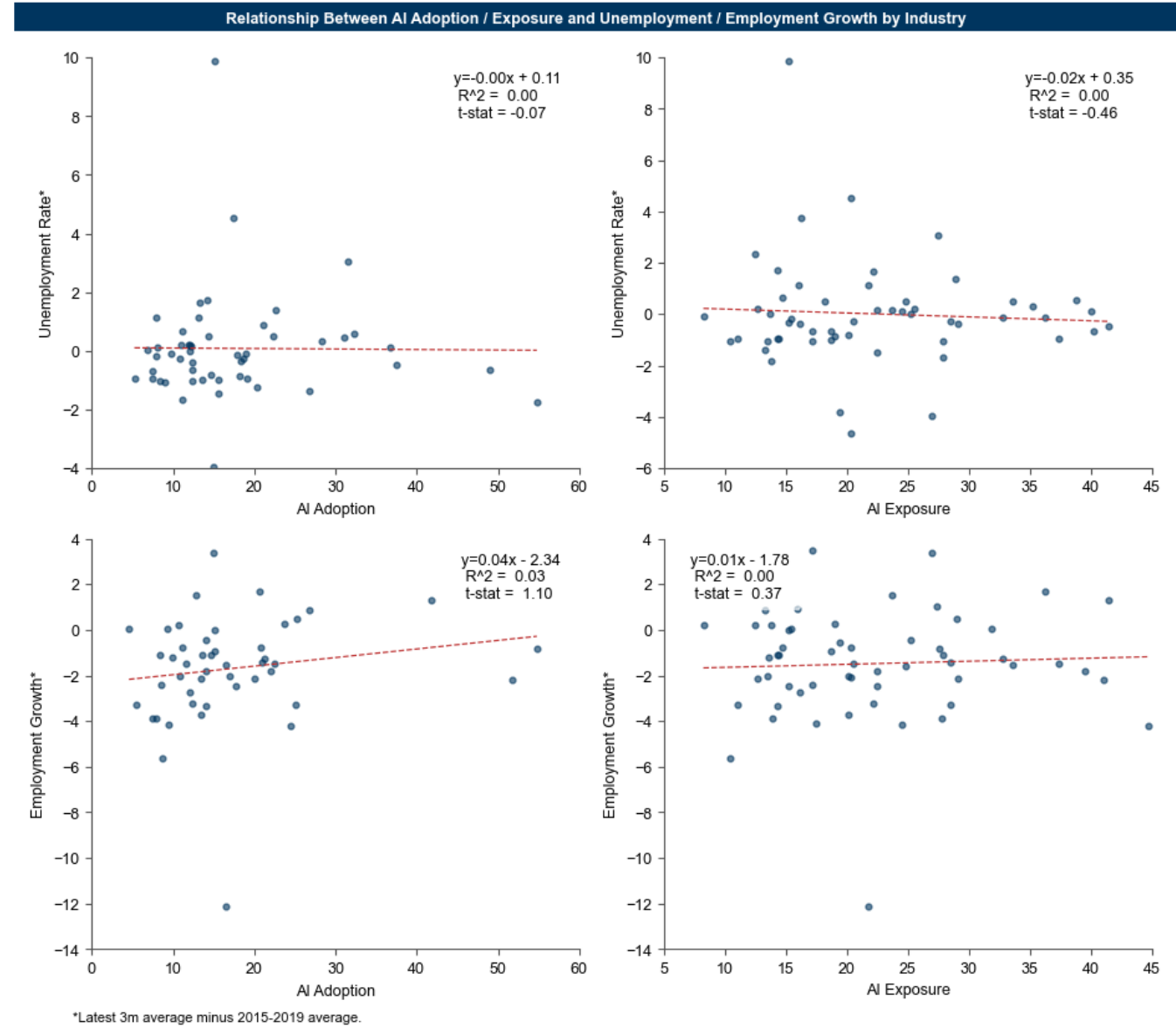
Exhibit 21: The Impact of AI Exposure and Adoption on Broad Labor Market Outcomes Remained Statistically Insignificant Through March



Note: All dependent variables are taken as the deviation from 2015–19 averages. Average coefficient over the past three months is plotted.

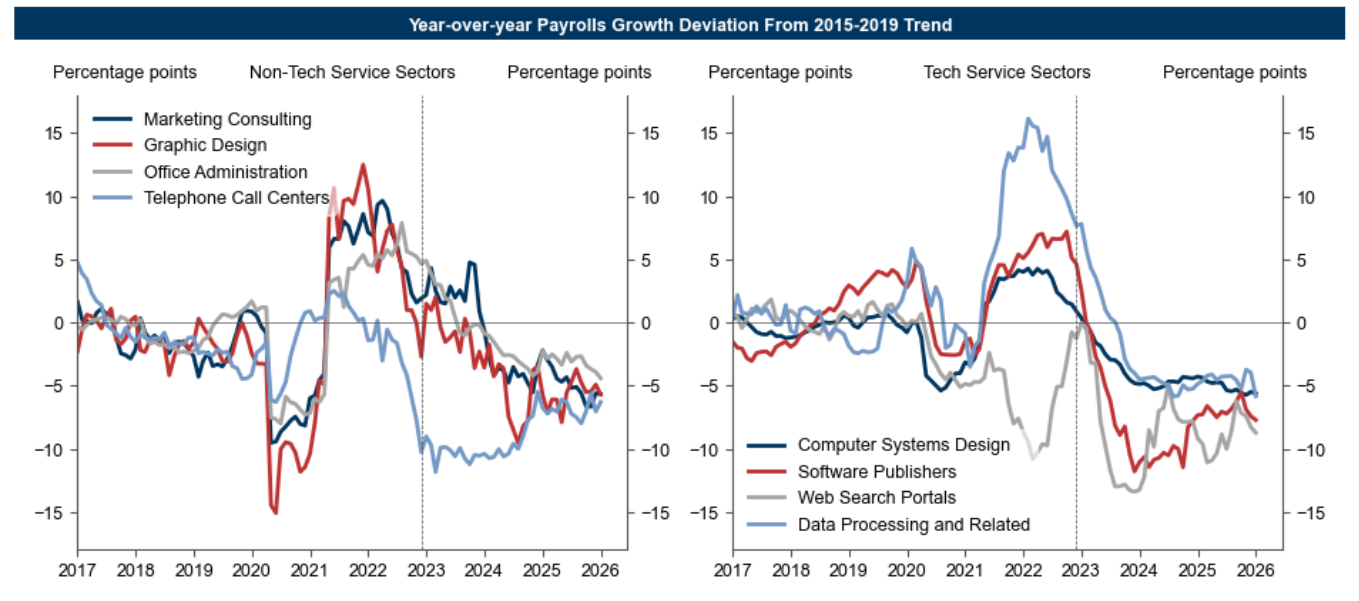
Source: IPUMS, US Bureau of Labor Statistics, Census Bureau, Goldman Sachs Global Investment Research

Exhibit 22: Still No Statistically Significant Relationship Between AI and Unemployment or Employment Growth



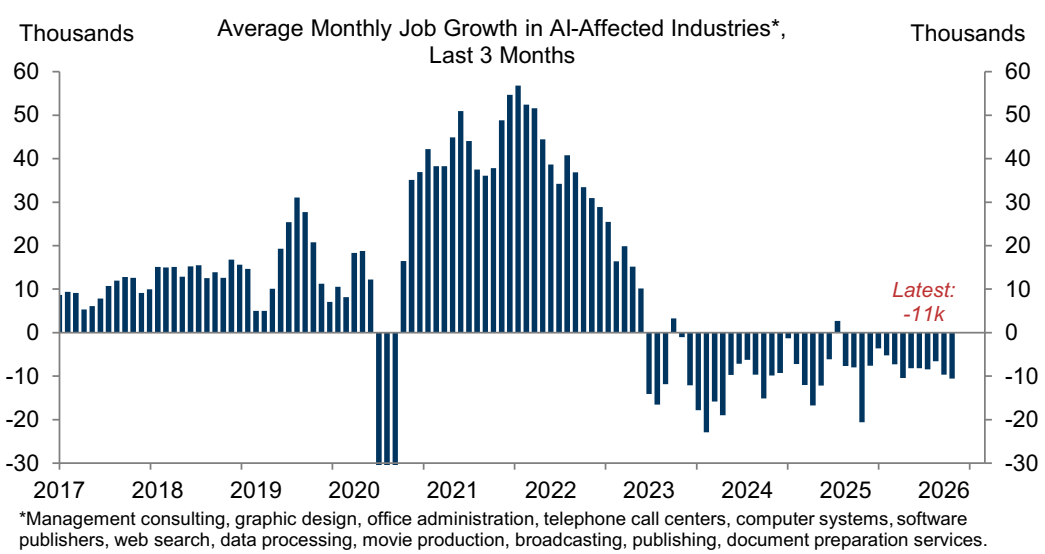
Source: US Bureau of Labor Statistics, IPUMS, Goldman Sachs Global Investment Research

Exhibit 23: Employment Growth Continues to Underperform in Occupations Where Anecdotes Suggest AI Is Having an Impact...



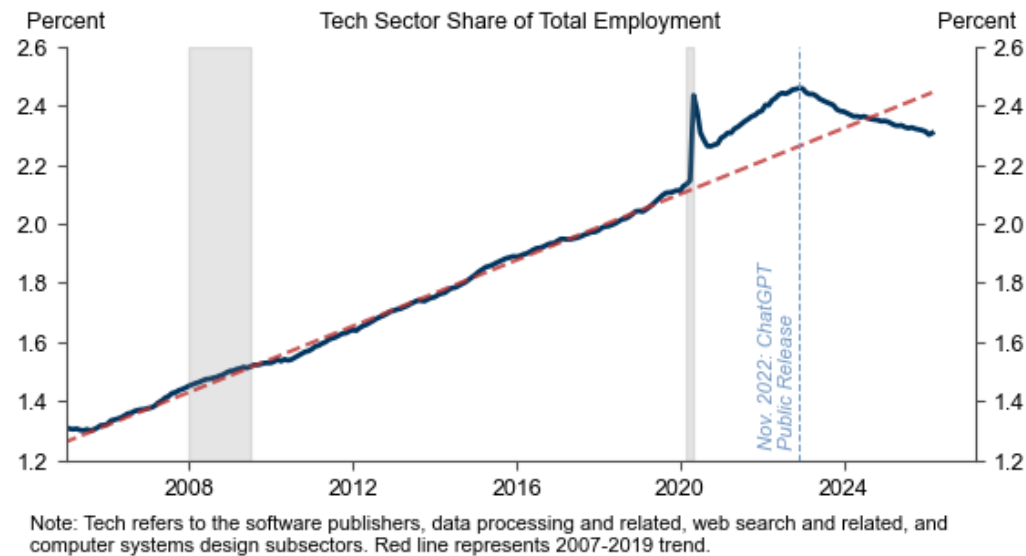
Source: US Bureau of Labor Statistics, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 24: ...With a 11k Overall Headwind to Job Growth From Industries Where Use Cases Have Been Established



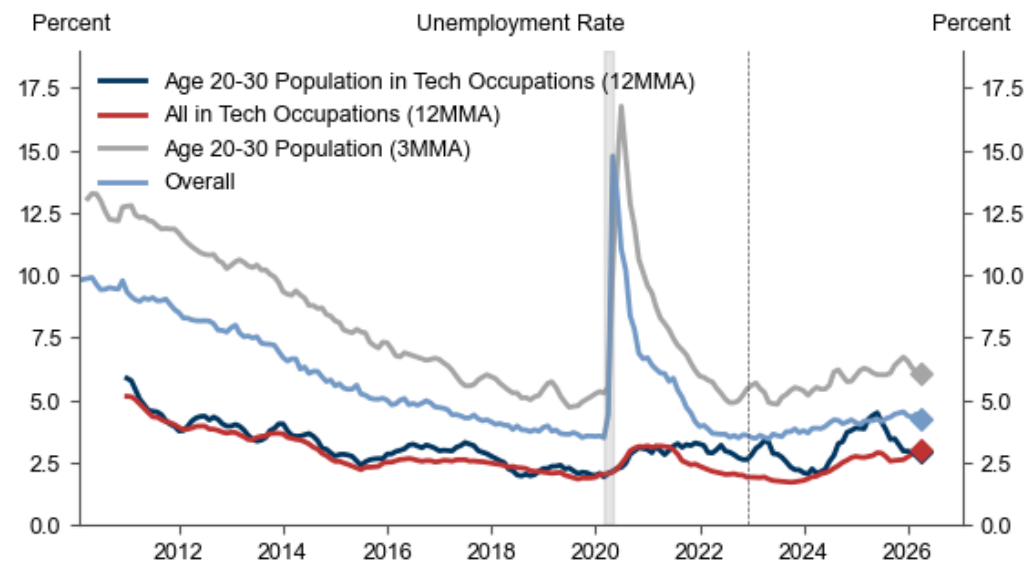
Source: US Bureau of Labor Statistics, Goldman Sachs Global Investment Research

Exhibit 25: Tech’s Employment Share Remains Low Relative to Its Long-run Trend



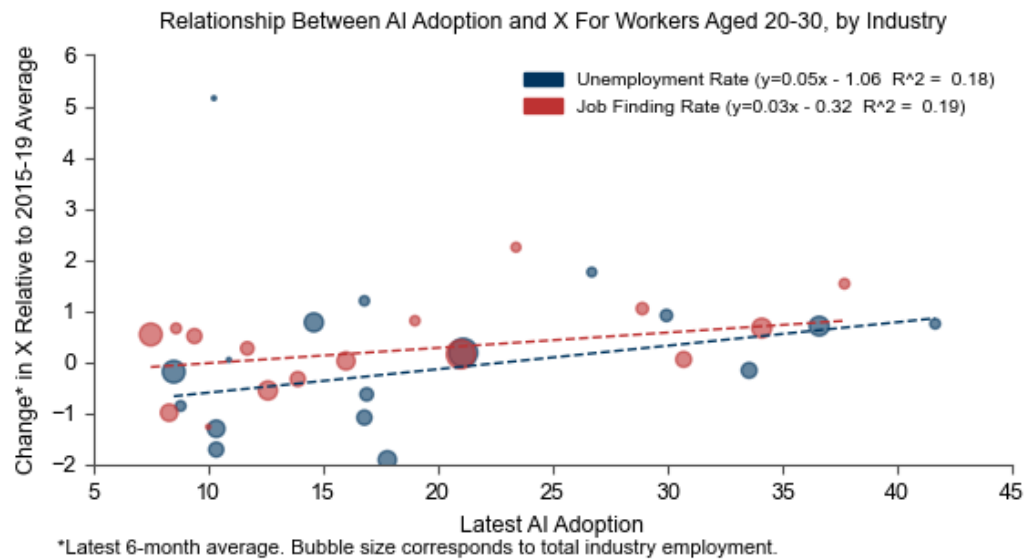
Source: US Bureau of Labor Statistics, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 26: Unemployment Among Young Tech Workers Has Fallen Back in Line With Unemployment for All Tech Workers



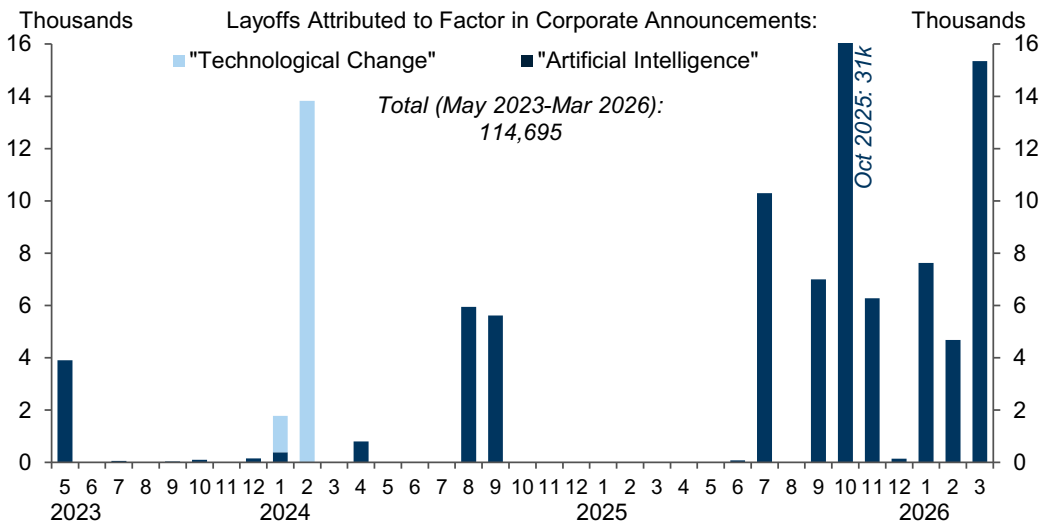
Source: US Bureau of Labor Statistics, IPUMS, Goldman Sachs Global Investment Research

Exhibit 27: We Continue to Observe a Slight Positive Correlation Between AI Adoption and Unemployment Among Younger Workers



Source: IPUMS, US Bureau of Labor Statistics, Census Bureau, Goldman Sachs Global Investment Research

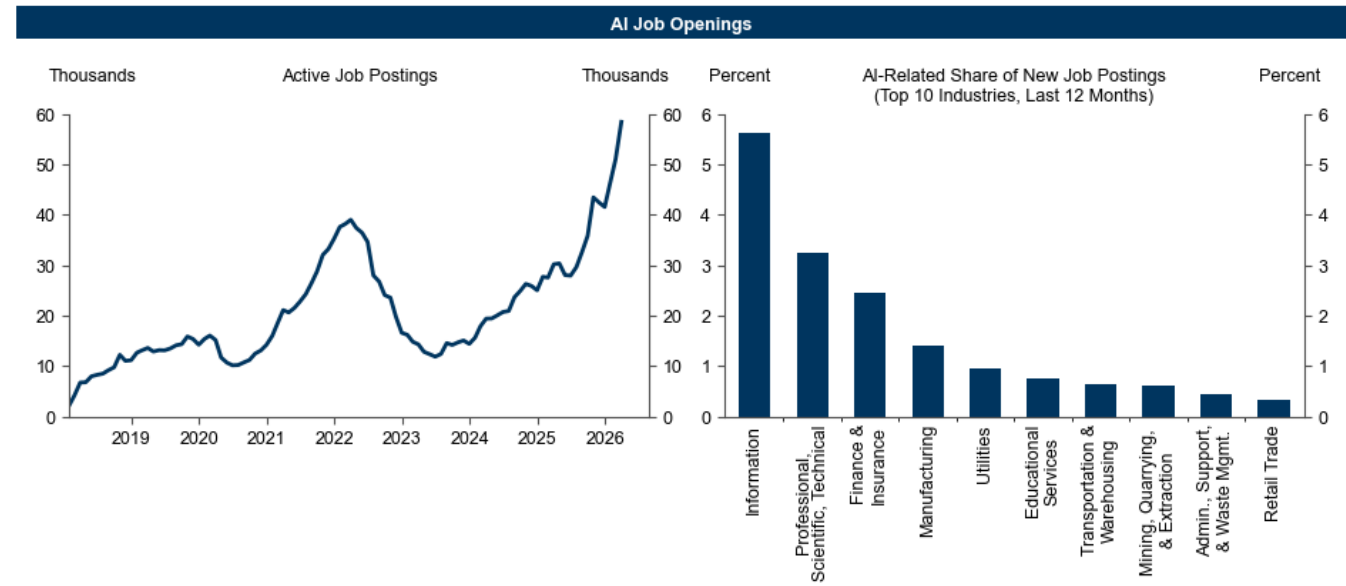
Exhibit 28: AI Was Cited in Layoff Announcements Affecting 15.3k Employees in March



Source: Challenger, Gray & Christmas, Goldman Sachs Global Investment Research

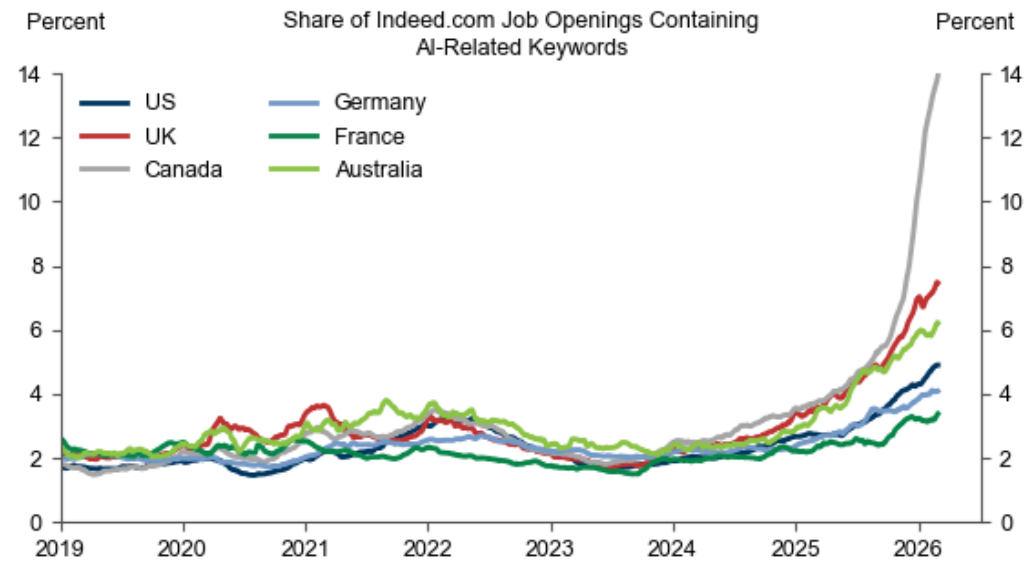
Labor Demand

Exhibit 29: AI Job Postings Remain Skewed Toward Information and Professional Services, as Well as Finance



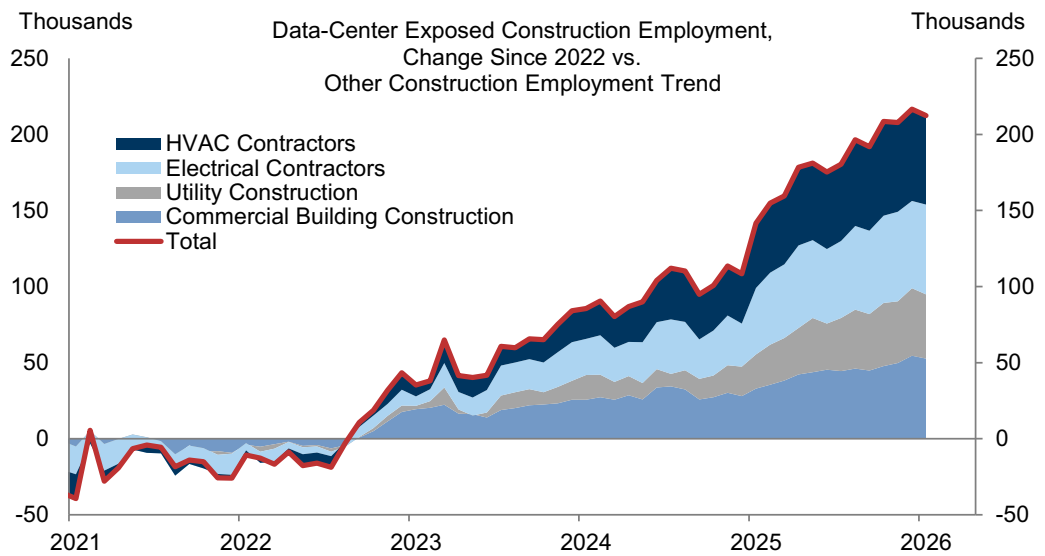
Source: University of Maryland, LinkUp, Goldman Sachs Global Investment Research

Exhibit 30: Share of AI-Related Job Postings Has Risen in Most DMs and Especially in Canada



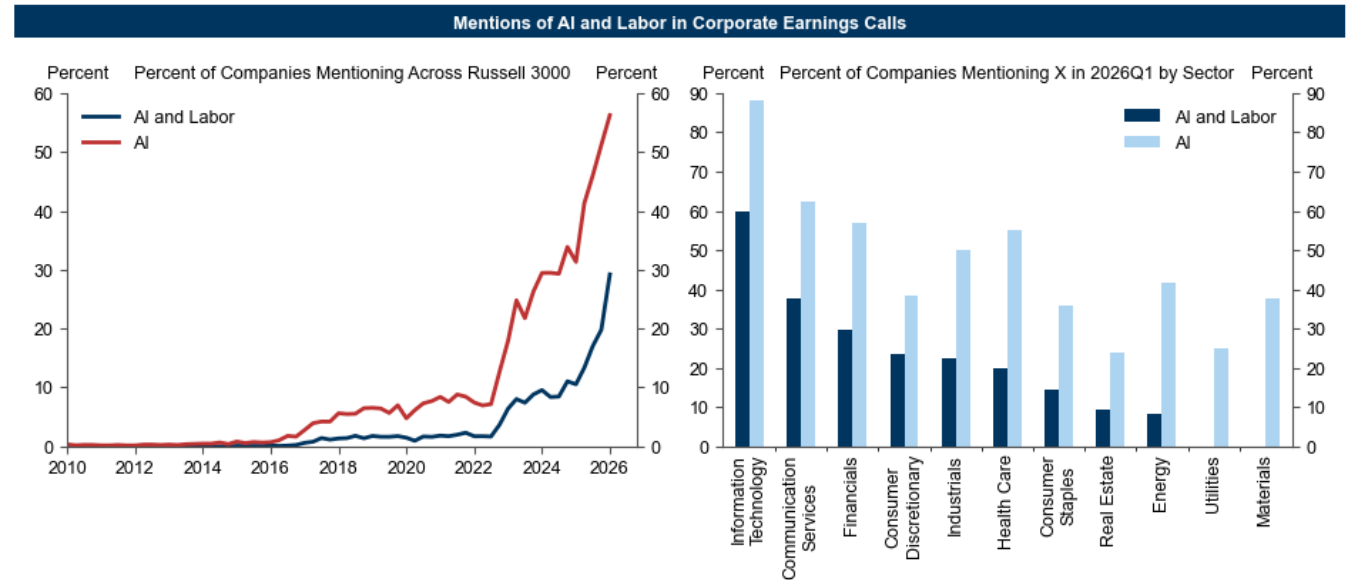
Source: Indeed Hiring Lab, Goldman Sachs Global Investment Research

Exhibit 31: Construction Employment in Data Center-Exposed Categories Has Increased by 212k Since 2022 (Relative to Broader Construction Employment Trends)



Source: US Bureau of Labor Statistics, Haver Analytics, Goldman Sachs Global Investment Research

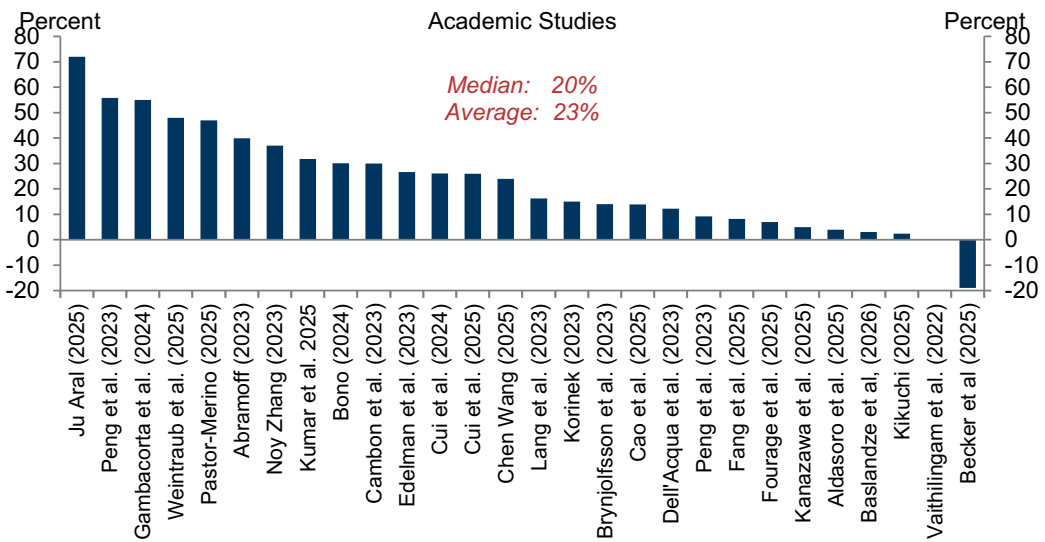
Exhibit 32: Nearly 60% of IT Companies Mentioned AI and Labor-Related Keywords in 2026Q1 Earnings Calls



Source: GS Data Works, Goldman Sachs Global Investment Research

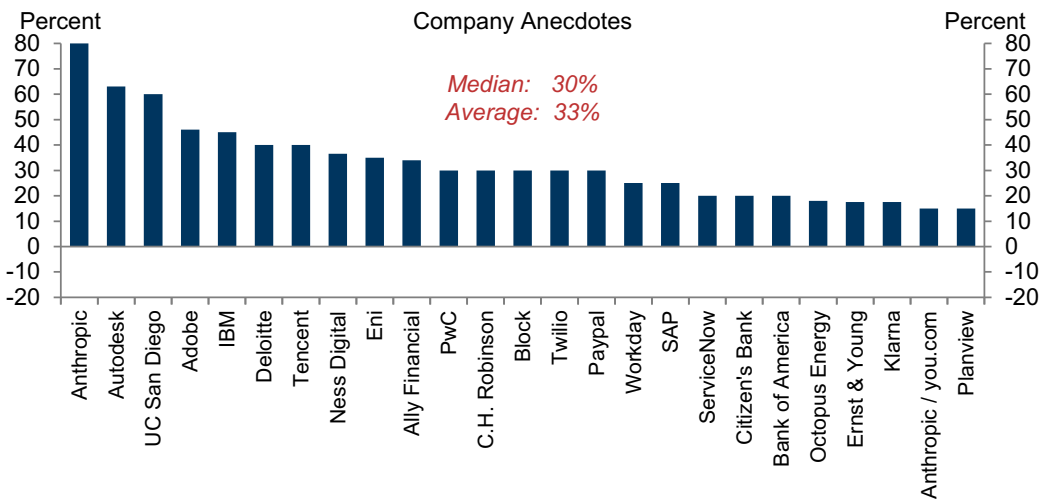
AI’s Impact on Productivity

Exhibit 33: Evidence From Academic Studies Suggests AI Adoption Boosts Labor Productivity by Around 23% on Average...



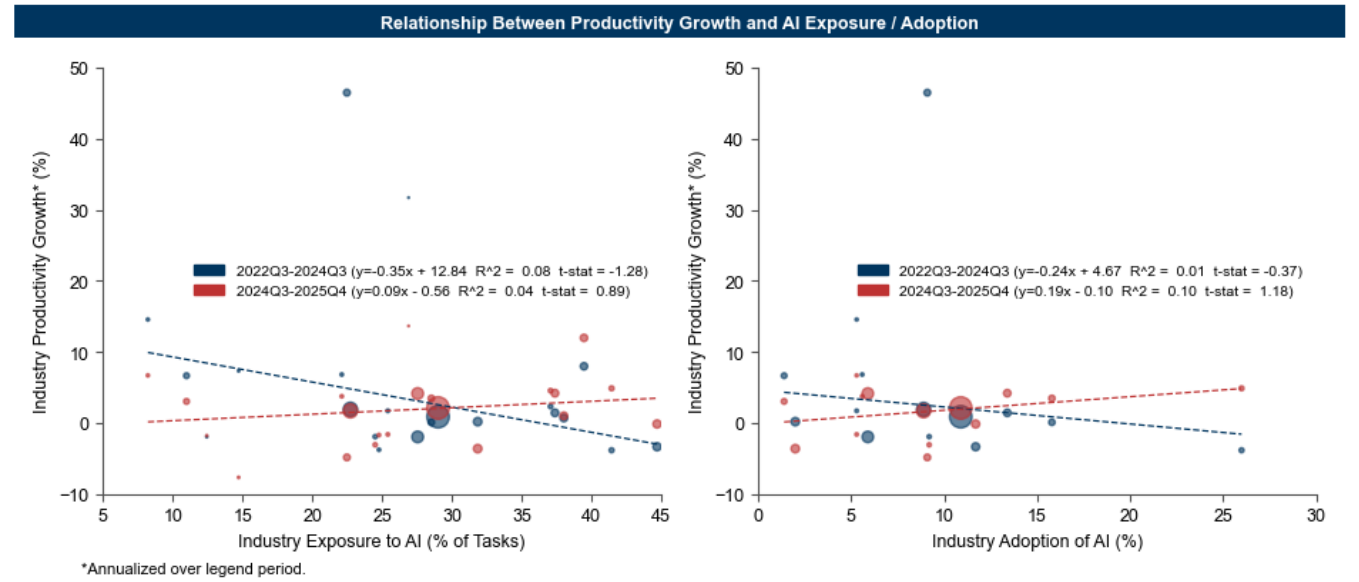
Source: Goldman Sachs Global Investment Research

Exhibit 34: ... And Company Anecdotes Suggest an Average Productivity Boost of 33%



Source: Company data, Data compiled by Goldman Sachs Global Investment Research

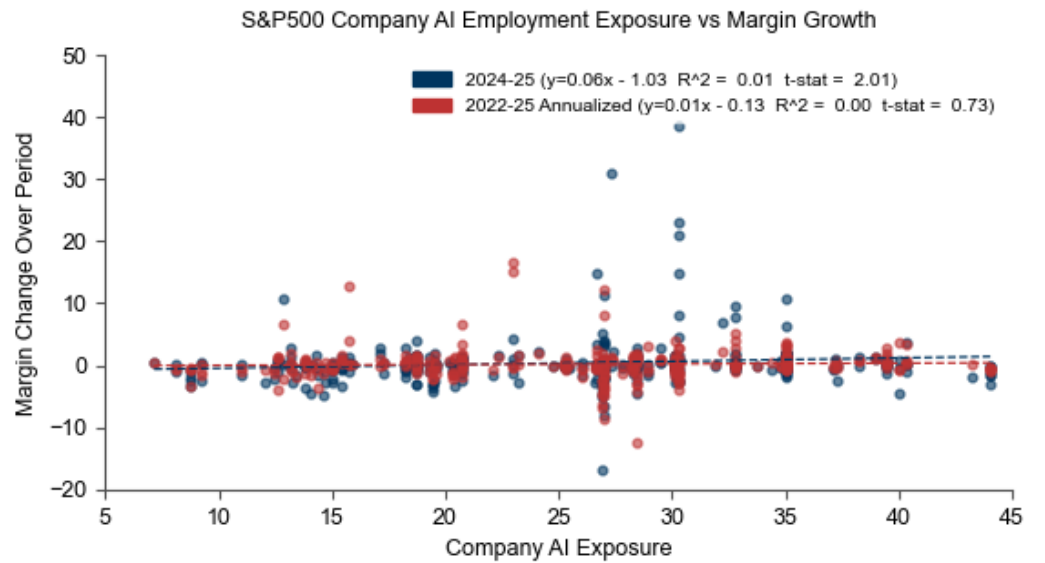
Exhibit 35: Industries with Higher Adoption rates Show Slightly Stronger Productivity Growth Over the Past Five Quarters



Note: Bubble size corresponds to total industry employment.

Source: US Bureau of Labor Statistics, Bureau of Economic Analysis, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 36: No Relationship Between Company Level AI Exposure and Margin Growth



Source: FactSet, Goldman Sachs Global Investment Research

Latest GS Research on Artificial Intelligence

- [Global Credit Trader: A closer look at AI-related debt supply](#), April 30
- [AI Spend vs. Benefit: Marking our views to market two years later](#), April 30
- [Credit Notes: AI-related debt issuance: The public vs. private mix](#), April 21
- [Global Strategy Views: The Technology Value Opportunity](#), April 7
- [US Daily: Breaking Down the Impact of AI on the Labor Market: Substitution vs. Augmentation](#), April 4
- For research from prior quarters, see the [Artificial Intelligence](#) page on our publishing portal.

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Disclosure Appendix

Reg AC

We, Sarah Dong and Joseph Briggs, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

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OIL ANALYST

How Fast Can Gulf Production Recover After Reopening?

- How fast can Gulf crude oil production, which we estimate to be down by 14.5mbd (or 57%) from pre-war levels ([Exhibit 1](#)), recover after the Strait of Hormuz reopens? We find that Gulf production is likely to mostly recover within a few months of reopening assuming 1) no renewed strikes on oil assets and 2) a full and safe reopening of the Strait in coming months. However, we see significant risks that the last leg of the recovery will take significantly longer and may not fully materialize, especially if the Strait were to remain closed for much longer.
- **Transportation and well flow rates are key.** Once the Strait safely re-opens, the key potential constraints on production will likely be the availability of 1) pipeline capacity and empty vessels to destock previously produced oil, and 2) of materials and workers for field workovers, and 3) well flow rates. We estimate that available empty tanker capacity in the Gulf has fallen by around 50% (130mb, #) since the start of the war. We also estimate historical peaks for flows through Hormuz of 23.3mb/d (vs. 20mb/d normal) and for pipeline redirection of 3.5mb/d above normal. Forced curtailments can lead to reservoir complexities in flow rates to varying degrees depending on the reservoir type, requiring intervention and workover jobs prior to opening up wells, to help restore prior production rates as much as possible ([Exhibit 2](#), quotes 1 and 2).
- **The longer the closure, the slower the production ramp.** The longer the Strait is effectively closed, the longer curtailments last, the slower the production recovery is likely to be ([Exhibit 2](#), quotes 2 and 3). This reflects potentially more work on wells, slower procurement of depleted inputs (e.g. drill pipes), and destocking constraining transport capacity for new production for longer.
- **The case for a solid recovery.** We see three reasons why Gulf production is likely to mostly recover within a few months after a full and safe reopening of the Strait:
 - a. The publicly reported evidence on physical damage to oil fields remains limited ([Exhibit 1](#)) in contrast to LNG assets.
 - b. Comments from the Saudi Aramco President and CEO in March suggested that Saudi production is likely to ramp up relatively quickly ([Exhibit 2](#), quote 5).
 - c. The history of supply disruptions ([Exhibit 5](#)) suggests that Saudi Arabia and

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the UAE are likely to deploy available spare capacity ([Exhibit 6](#)) to stabilize markets.

- **Risks to long-run production.** We see three reasons why a full recovery may take several quarters and risks that the recovery may be only partial after a prolonged closure.
 - a. There are large differences across:
 - Gulf fields in terms of reservoir characteristics, infrastructure sophistication, maintenance, and sanctions risks (e.g. Iran). For instance, the share of country production with relatively low reservoir pressure is estimated to be higher in Iran and Iraq than elsewhere in the Gulf ([Exhibit 7](#)).
 - Historical episodes in terms of the speed and extent of production recovery (partly reflecting the differences in causes of curtailments, and recognizing that today's shock is unprecedented) ([Exhibit 8](#)).
 - b. An average across external forecasts of Persian Gulf production (incl. from EIA, IEA) show recoveries as % of lost production of 70% after 3 months of reopening and of 88% after 6 months ([Exhibit 9](#)).
 - c. While not our baseline, the risk of large scarring to oil production capacity (we documented for several of the previous five largest oil supply shocks) may pick up if hostilities were to resume.

How Fast Can Gulf Production Recover After Reopening?

Exhibit 1: We Estimate 14.5mb/d of Persian Gulf Crude Production Curtailed in April, Primarily on Precautionary and Stock Management Measures

Estimated Gulf Crude Production in April 2026 (mb/d)				
	Estimated Crude Production Without War	Still Producing	Not Producing Because of Reported Physical Damage	Not Producing for Other Reasons
Total	25.4	11.0	0.3	14.2
Saudi Arabia	10.1	6.1	0.3	3.7
Iraq	4.4	1.4	0	3.0
Iran	3.5	1.0	0	2.5
UAE	3.5	0.8	0	2.7
Kuwait	2.6	1.4	0	1.2
Qatar	1.3	0.2	0	1.0

Under “Estimated Crude Production Without War” we report April 2026 production projections from our GS February 2026 balance. Argus reported on 12 April 2026 that Saudi Arabia’s Energy Ministry said repair work was ongoing at Khurais field, where production capacity had been reduced by about 0.3mb/d, following a strike on one of its East-West pipeline pumping stations. The Ministry said that full restoration would be announced once repairs were completed.

Source: IEA, Goldman Sachs Global Investment Research

Exhibit 2: Insufficient Reservoir Pressure/Flow Rates Following Forced Curtailments May Be a More Persistent Constraint on Production of Some Wells

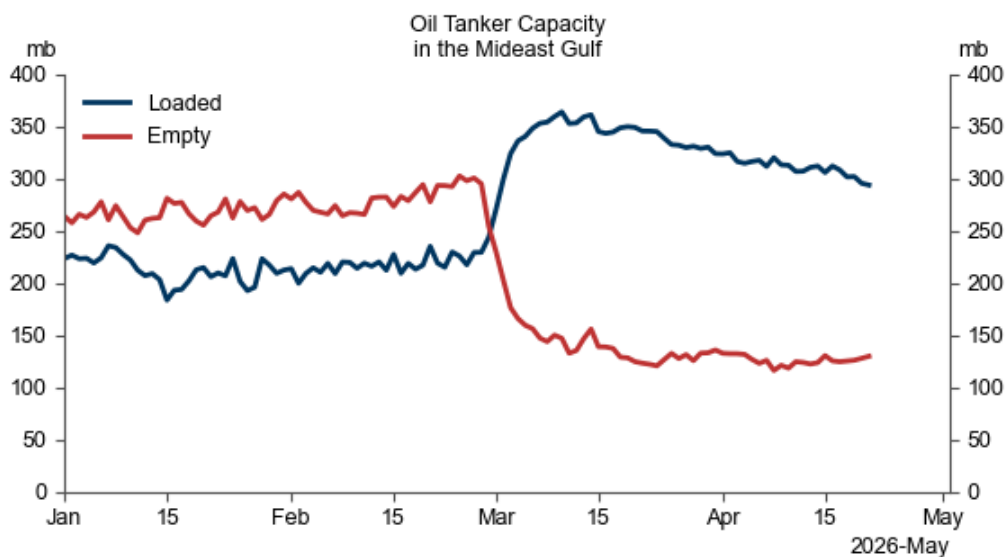
Prospects for Recovery in Upstream Oil Production			
Quote Number	Date	Speaker	Speed of Recovery
1	22-Apr	Girish K. Saligram (CEO of Weatherford International)	When you’ve also got production that shut in [...] it’s not a guarantee that you’re going to get back at the exact same flow rates .
2	21-Apr	Shannon Slocum (Executive Vice President and Chief Operating Officer of Halliburton)	I’d say the longer they [wells] get shut in, the more complex that gets .
3	21-Apr	Jeffrey A. Miller (Chairman of the Board, President and Chief Executive Officer of Halliburton)	The longer things are shut in, typically, the more complex they are to bring back on .
4	24-Mar	Sheikh Nawaf (CEO of Kuwait Petroleum Corporation)	Kuwait would take three to four months to return output to full production levels even if the war were to end today.
5	10-Mar	Amin Nasser (President and CEO of Saudi Aramco)	We can ramp up in days , and not weeks for sure and, to reach whatever, when we talk about, we have an ample spare capacity as well, and in certain areas, we use restriction on wells rather than shutdowns.

Source: Reuters, Saudi Aramco, Halliburton, Weatherford International, Goldman Sachs Global Investment Research

Production Process

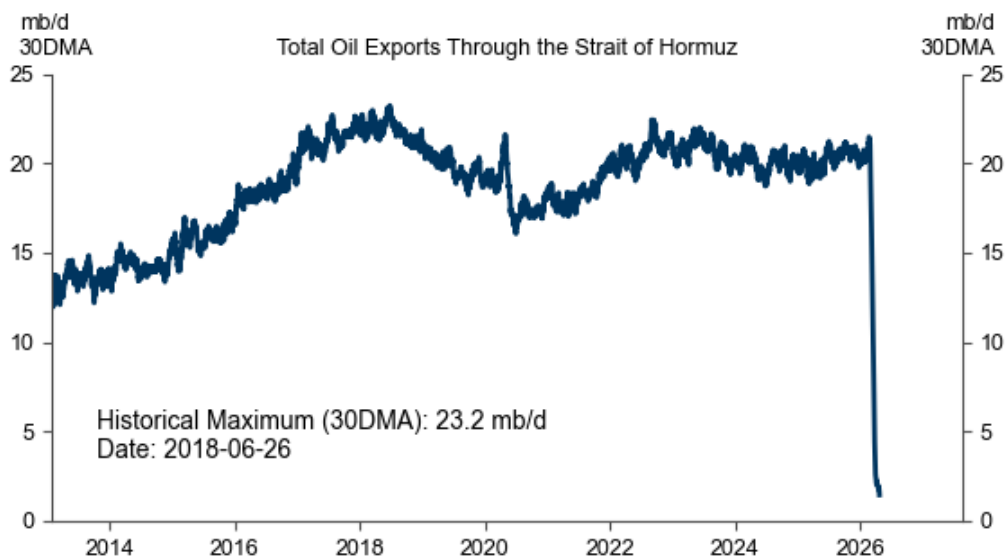
Transportation and Well Flow Rates Are Key Potential Constraints

Exhibit 3: We Estimate that Available Empty Tanker Capacity in the Persian Gulf Has Fallen by Around 130mb (Around 50%)



Source: Kpler, Goldman Sachs Global Investment Research

Exhibit 4: Oil Flows Through the Strait of Hormuz Exceeded Their Normal 20mb/d Pace by Over 3mb/d in mid-2018



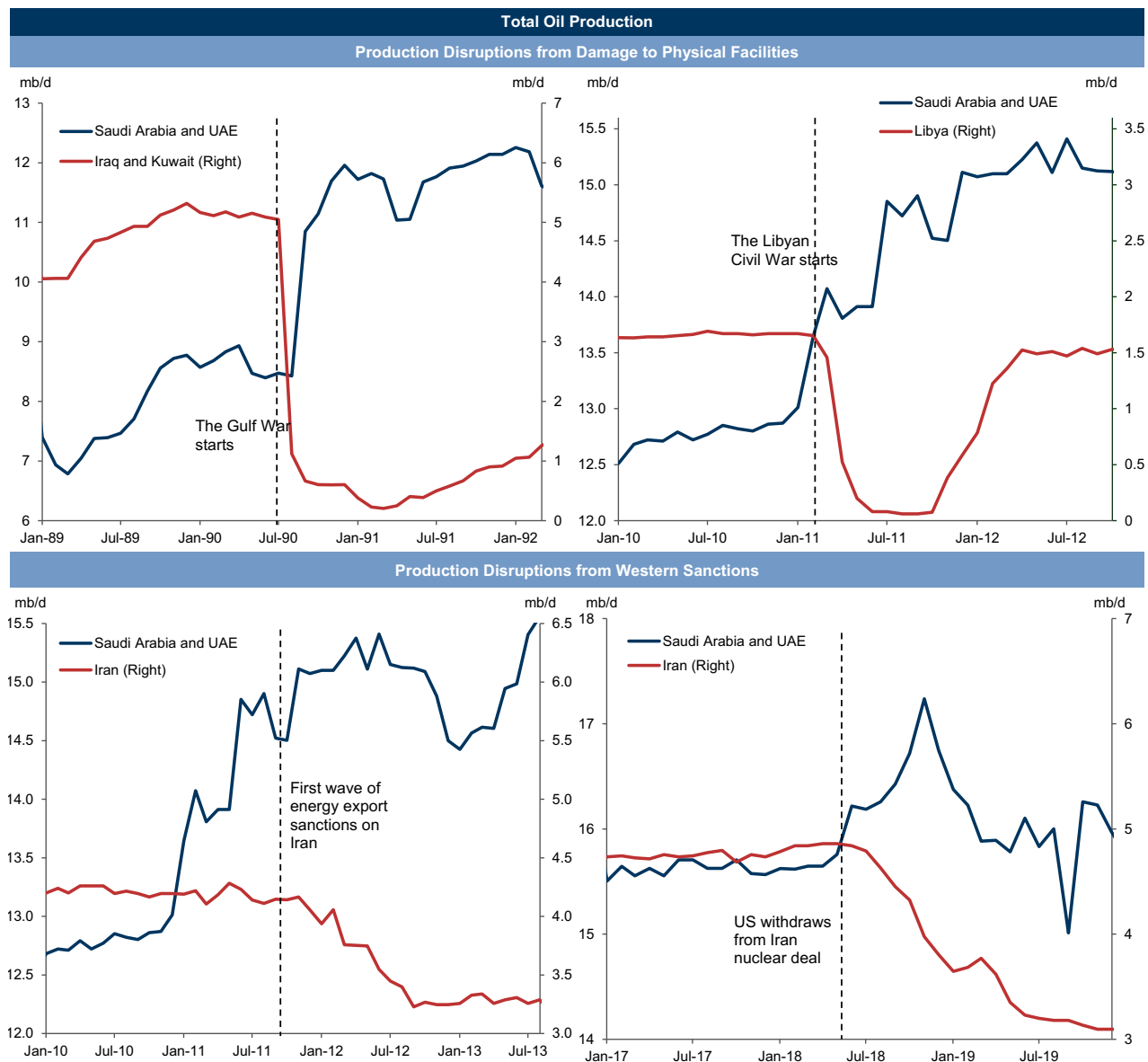
Oil exports includes crude, NGLs, and refined products.

Source: Kpler, Goldman Sachs Global Investment Research

The Case for a Solid Recovery

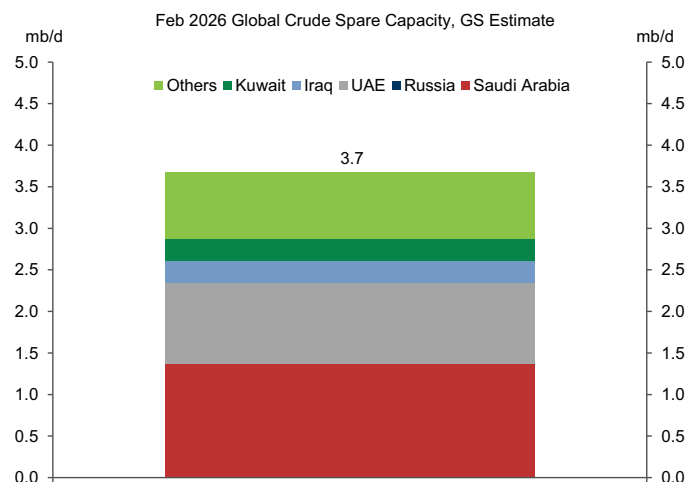
Core OPEC Market Stabilization

Exhibit 5: Saudi Arabia and UAE Production Historically Tends to (Eventually) Offset Much of Disrupted Supply



Source: IEA, Goldman Sachs Global Investment Research

Exhibit 6: We Estimate That Saudi Arabia and UAE Could Deploy Over 2mb/d of Spare Production vs. Pre-War Levels to Stabilize Markets After a Safe and Full Reopening

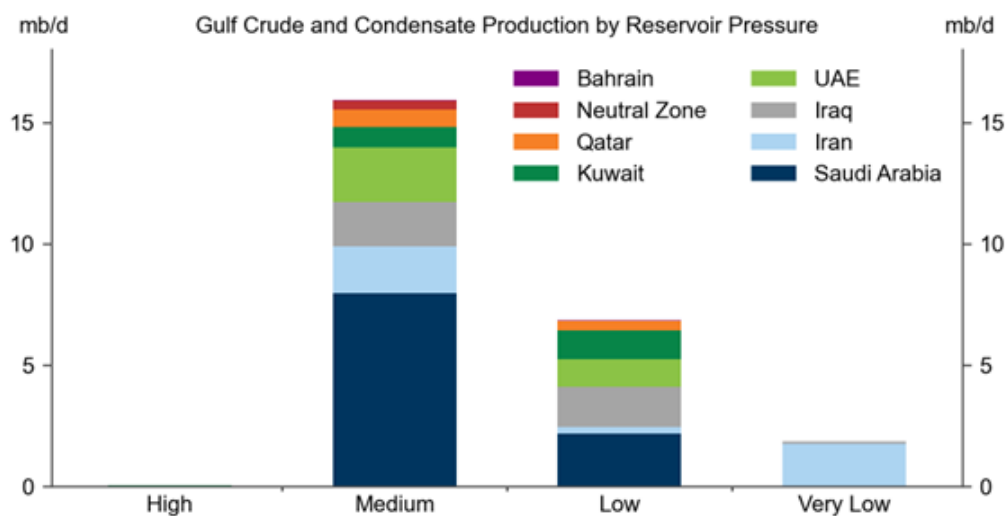


Source: IEA, EIA, Goldman Sachs Global Investment Research

Risks to Long-Run Production

Differences Across Fields and Countries

Exhibit 7: Most of Gulf Reservoirs Are Estimated to Have Medium Pressure, But Nearly Half of Iranian Reservoirs Are Estimated to Have Low Pressure



Source: Rystad Energy

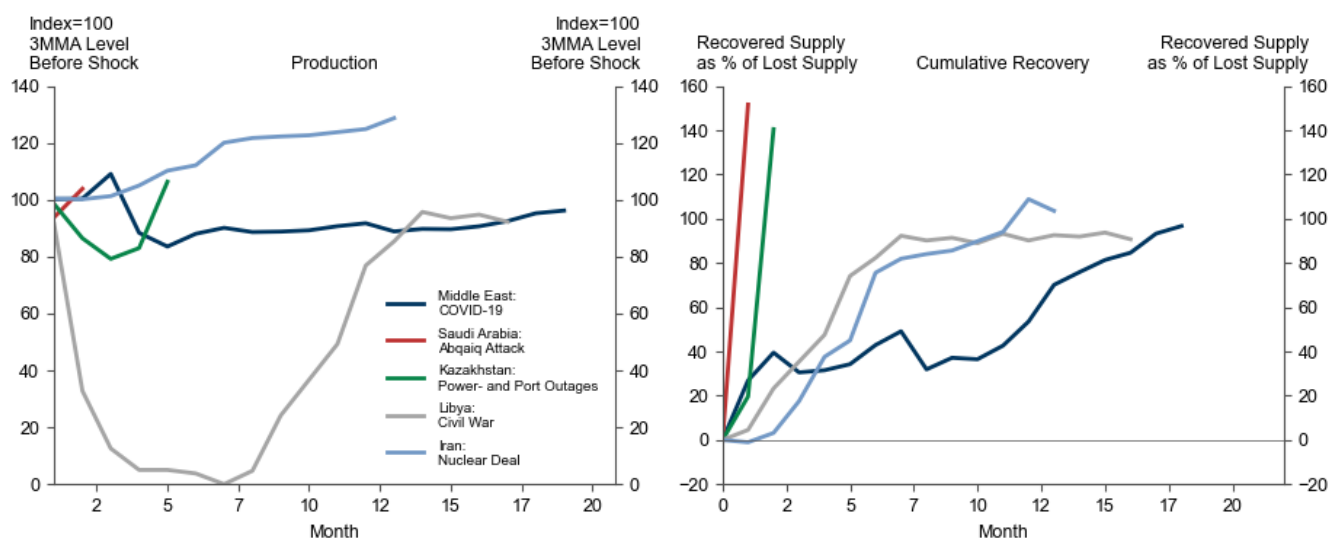
Production is a 2026 annual average projection. Pressure is not a sufficient statistic to predict workovers on wells. Other factors (e.g. sand granularity) play a role too.

Source: Rystad Energy, IEA, Goldman Sachs Global Investment Research

Differences Across Historical Production Recovery Episodes

Exhibit 8: While the Hormuz Shock Is Unprecedented, Our Analysis of Historical Production Recovery Episodes Shows Large Differences in Recovery Paces (Partly Reflecting Different Causes of Curtailments)

Differences Across Historical Crude Production Recovery Episodes



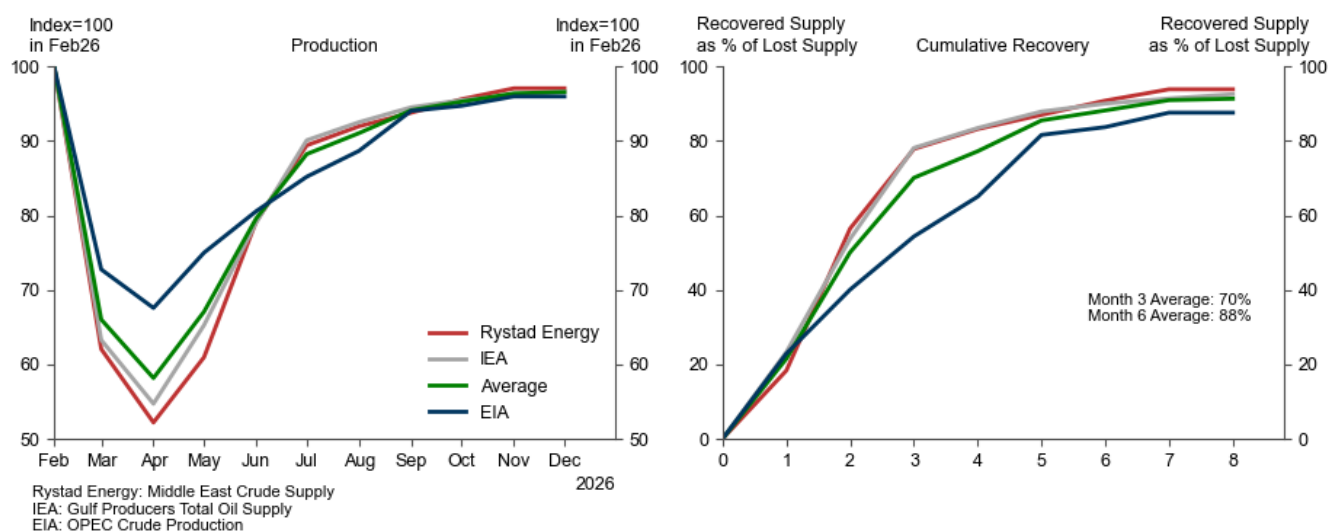
We consider crude production data.

Source: IEA, Goldman Sachs Global Investment Research

Forecasts From Energy Agencies

Exhibit 9: An average Across External Forecasts of Persian Gulf Production in 2026 Shows Recoveries as % of Lost Production of 70% After 3 Months of Reopening and of 88% After 6 Months

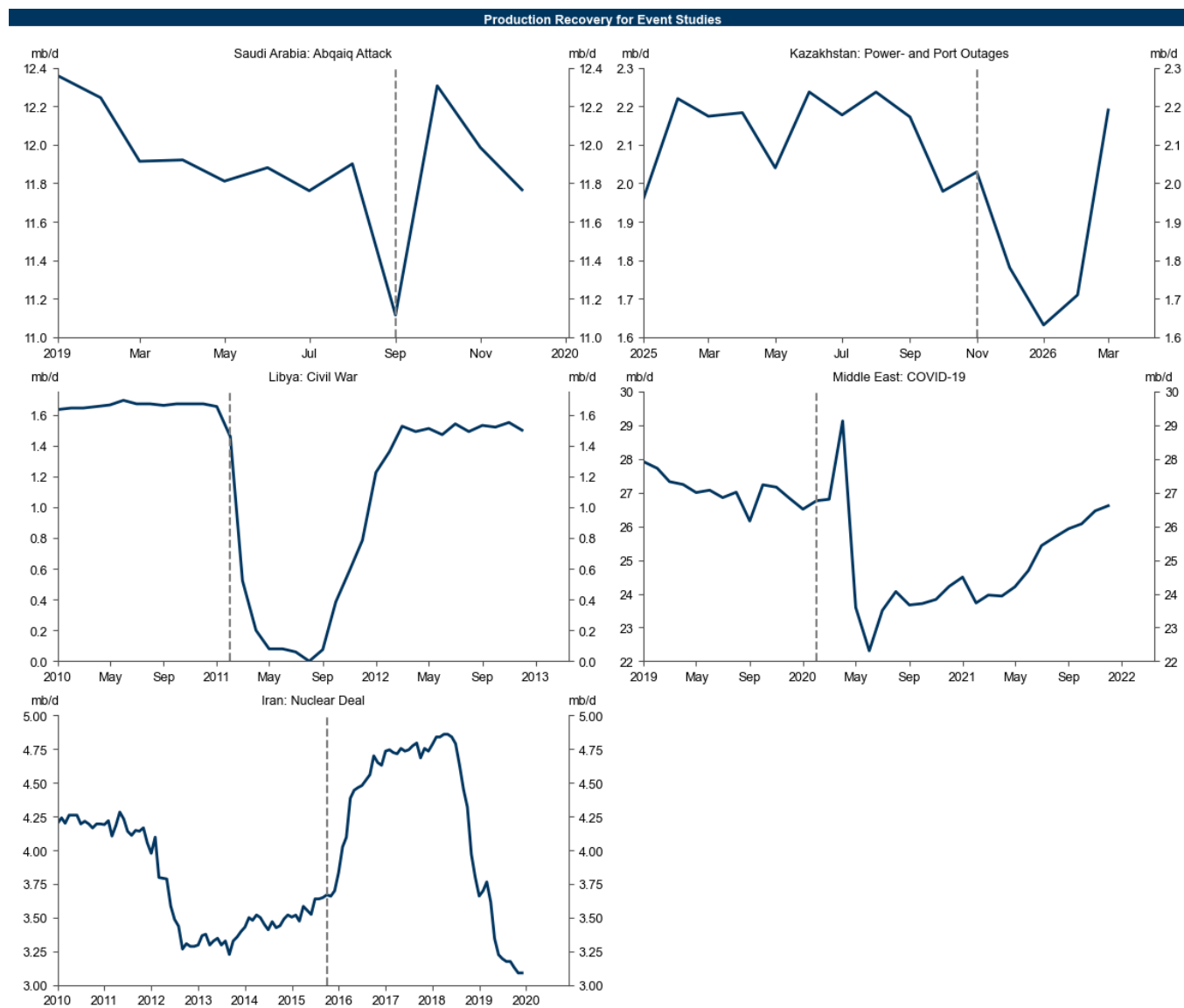
Forecasts Comparison Across Agencies



Source: Rystad Energy, IEA, EIA, Goldman Sachs Global Investment Research

Appendix

Exhibit 10: We Look at Five Historical Episodes of Production Disruptions and Recovery But Ramp-Up Patterns Differ Across the Shocks (Which All Differ in Cause and Nature)



Source: IEA, Goldman Sachs Global Investment Research

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